Properties of Triangle

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Properties of Triangle

Introduction

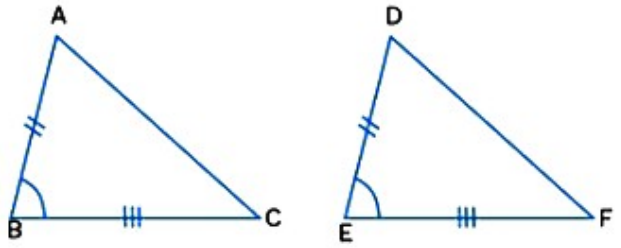
Congruence of Triangles:

If ΔABC is congruent to ΔDEF, we write $\Delta ABC \cong \Delta DEF$. This happens when

$AB = DE$, $BC = EF$, $AC = DF$ and $\angle A = \angle D$, $\angle B = \angle E$, $\angle C = \angle F$

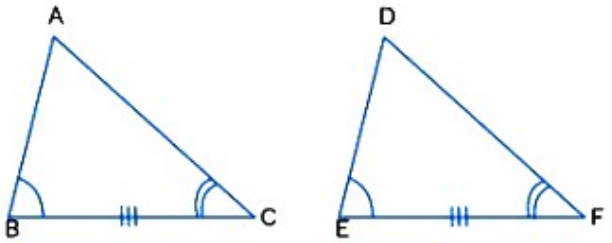
1. Side-Angle-Side (SAS) Congruence criterion:

In ΔABC and ΔDEF if
$AB = DE$
$\angle ABC = \angle DEF$
$BC = EF$ then
$\Delta ABC \cong \Delta DEF$



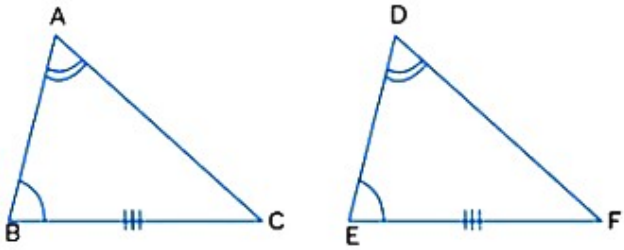
2. Angle-Side-Angle (ASA) Congruence criterion:

In ΔABC and ΔDEF if
$\angle ABC = \angle DEF$
$BC = EF$
$\angle BCA = \angle EFD$ then
$\Delta ABC \cong \Delta DEF$



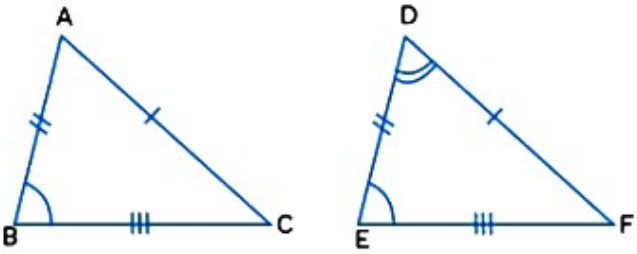
3. Angle-Angle-Side (AAS) Congruence criterion:

In ΔABC and ΔDEF if
$\angle CAB = \angle FDE$
$\angle ABC = \angle DEF$
$BC = EF$ then
$\Delta ABC \cong \Delta DEF$

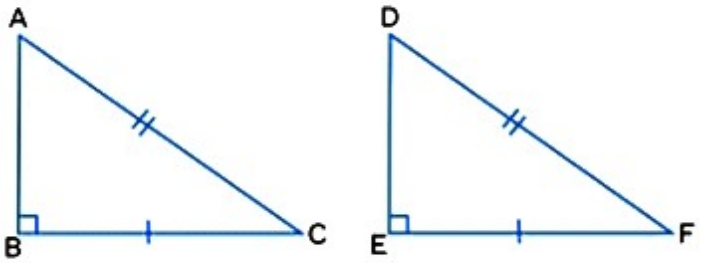


4. Side-Side-Side (SSS) Congruence criterion:

In ΔABC and ΔDEF if
$AB = DE$
$BC = EF$
$AC = DF$ then
$\Delta ABC \cong \Delta DEF$



5. Right Angle-Hypotenuse-Side (RHS) Congruence criterion:



In ΔABC and ΔDEF if

$\angle ABC = \angle DEF = \frac{\pi}{2}$

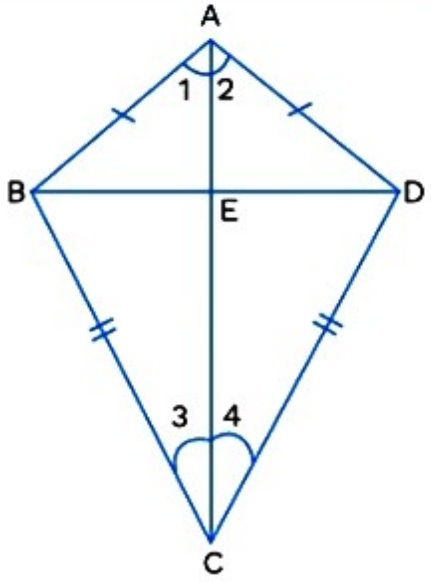
$AC = DF$

$BC = EF$ then

$\Delta ABC \cong \Delta DEF$

Q. **In the figure ABCD is a quadrilateral in which AB = AD and BC = DC. Prove that:**
(i) AC bisects each of the angle A and C.
(ii) BE = ED
(iii) $\angle ABC = \angle ADC$.
Can we say that AE = EC?

Sol.



In ΔABC & ΔADC
$AB = AD$ (Given)
$BC = DC$ (Given)
$AC = AC$ (common side)
Hence by SSS, $\Delta ABC \cong \Delta ADC$
which implies $\angle 1 = \angle 2$ and $\angle 3 = \angle 4$
and $\angle ABC = \angle ADC$
Hence AC bisects each of the angle A and C
Now in ΔABE and ΔADE
$AB = AD$ (Given)
$\angle 1 = \angle 2$ (Proved)
$AE = AE$ (Common)
Hence by SAS, $\Delta ABE \cong \Delta ADE$
Which implies $BE = ED$
We cannot say that AE = EC

About Triangle:

In any triangle the three sides and the three angles are generally called the elements of triangle. In any triangle ABC, the measures of the angles BAC, CBA and ACB are denoted by the letter A, B and C respectively. The side BC, CA and AB are opposite to the angles A, B and C respectively are denotes by a, b and c. These six elements of a triangle are not independent and are connected by the relations:

(i) $A + B + C = \pi$

(ii) $a + b > c, b + c > a, c + a > b$

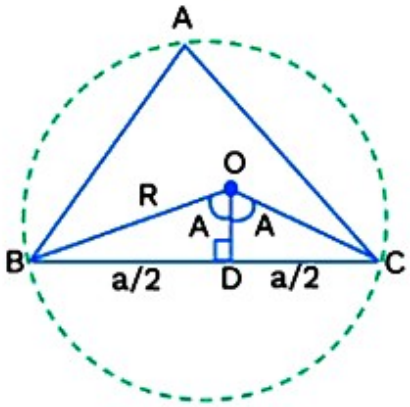
(iii) $|a - b| < c, |b - c| < a, |c - a| < b$

Sine Formula:

In any triangle ABC, the ratios of the sides to sine of the opposite angles are equal.

i.e. $\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = 2R$

where R is circumradius of ΔABC



Note: $\Delta = \frac{1}{2}ab\sin C$

$\Delta = \frac{1}{2}bc\sin A$, $\Delta = \frac{1}{2}ca\sin B$

Proof: In ΔBOD

$$\sin A = \frac{\frac{a}{2}}{R}$$

$$\Rightarrow 2R = \frac{a}{\sin A}$$

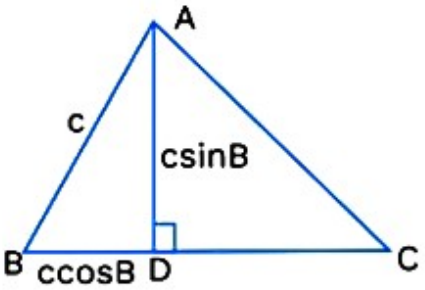
Similarly, others can be proved

Now area of triangle

$$\Delta = \frac{1}{2}(BC)(AD)$$

$$= \frac{1}{2}(a)(c\sin B)$$

$$\Delta = \frac{1}{2}ac\sin B$$



Similarly, others can also be proved.

Napier's Formula:

(i) $\tan\left(\frac{B-C}{2}\right) = \frac{b-c}{b+c}\cot\frac{A}{2}$

(ii) $\tan\left(\frac{C-A}{2}\right) = \frac{c-a}{c+a}\cot\frac{B}{2}$

(iii) $\tan\left(\frac{A-B}{2}\right) = \frac{a-b}{a+b}\cot\frac{C}{2}$

Proof: $\frac{b-c}{b+c} = \frac{2R\sin B - 2R\sin C}{2R\sin B + 2R\sin C}$

$$= \frac{2\cos\left(\frac{B+C}{2}\right)\sin\left(\frac{B-C}{2}\right)}{2\sin\left(\frac{B+C}{2}\right)\cos\left(\frac{B-C}{2}\right)}$$

$$= \frac{\sin\left(\frac{A}{2}\right)}{\cos\left(\frac{A}{2}\right)}\tan\left(\frac{B-C}{2}\right)$$

$$\Rightarrow \quad \tan\left(\frac{B-C}{2}\right) = \frac{b-c}{b+c}\cot\frac{A}{2}$$

Similarly, others can also be proved.

Cosine Formula:

$$\cos A = \frac{b^2+c^2-a^2}{2bc},\ \cos B = \frac{a^2+c^2-b^2}{2ac},\ \cos C = \frac{a^2+b^2-c^2}{2ab}$$

$\cos A > 0 \Leftrightarrow \angle A$ is acute $\Leftrightarrow b^2 + c^2 - a^2 > 0 \Leftrightarrow b^2 + c^2 > a^2$

$\cos A < 0 \Leftrightarrow \angle A$ is obtuse $\Leftrightarrow b^2 + c^2 < a^2$

$\cos A = 0 \Leftrightarrow \angle A = \frac{\pi}{2} \Leftrightarrow b^2 + c^2 = a^2$

Proof: In ΔADC

$\Rightarrow \quad AD^2 + DC^2 = AC^2$

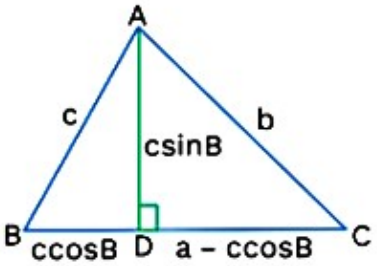
$\Rightarrow \quad (c\sin B)^2 + (a - c\cos B)^2 = b^2$

$\Rightarrow \quad c^2\sin^2 B + a^2 + c^2\cos^2 B - 2ac\cos B = b^2$

$\Rightarrow \quad a^2 + c^2 - b^2 = 2ac\cos B$

$\Rightarrow \quad \cos B = \frac{a^2+c^2-b^2}{2ac}$

Similarly, others can also be proved.



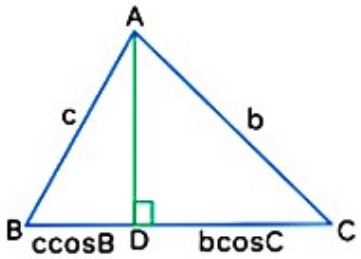
Uses:

To solve the triangle, if two sides and the included angle (e.g. b, c, $\angle A$) are given.

To solve the triangle, if two sides and any one angle (e.g. a, b, $\angle A$) are given.

Projection Formula:

$a = c\cos B + b\cos C$
$b = a\cos C + c\cos A$
$c = a\cos B + b\cos A$



Proof: In ΔABC

$BC = BD + DC$

$a = c\cos B + b\cos C$

Similarly, others can also be proved.

Q. **In a Δ, sides are proportional to $\cos\frac{A}{2}, \cos\frac{B}{2}, \cos\frac{C}{2}$. Find angles of triangle where A, B and C are angles of triangle.**

Sol. Given

$$\frac{a}{\cos\frac{A}{2}} = \frac{b}{\cos\frac{B}{2}} = \frac{c}{\cos\frac{C}{2}} \quad \ldots(1)$$

$$\frac{2R\sin A}{\cos\frac{A}{2}} = \frac{2R\sin B}{\cos\frac{B}{2}} = \frac{2R\sin C}{\cos\frac{C}{2}} \quad \text{(by sine-rule)}$$

$$\Rightarrow \sin\frac{A}{2} = \sin\frac{B}{2} = \sin\frac{C}{2}$$

$$\Rightarrow \frac{A}{2} = \frac{B}{2} = \frac{C}{2}$$

$$\Rightarrow A = B = C = \frac{\pi}{3}$$

Q. **In a triangle ABC, a = 5, b = 7 and $\sin A = \frac{3}{4}$. How many such triangles are possible?**

Sol. By sine rule

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$\Rightarrow \frac{5}{\frac{3}{4}} = \frac{7}{\sin B} = \frac{c}{\sin C}$$

$$\Rightarrow \sin B = \frac{21}{20} \text{ (which is not possible)}$$

Hence no such triangle is possible.

Q. **If the angles of a triangle be in the ratio 1 : 2 : 7, then the ratio of its greatest side to the least side is ?**

Sol. Let A : B : C = 1 : 2 : 7

$A = \theta, B = 2\theta, C = 7\theta$

now $A + B + C = \pi \Rightarrow 10\theta = \pi \Rightarrow \theta = \frac{\pi}{10}$

$\because$ C is greatest angle $\Rightarrow$ c is greatest

A is least angle $\Rightarrow$ a is least

Now $\frac{a}{\sin A} = \frac{c}{\sin C}$

$\Rightarrow \frac{c}{a} = \frac{\sin C}{\sin A}$

$$\frac{c}{a} = \frac{\sin\frac{7\pi}{10}}{\sin\frac{\pi}{10}} = \frac{\sin\frac{3\pi}{10}}{\sin\frac{\pi}{10}} = \frac{\sqrt{5}+1}{4} \times \frac{4}{\sqrt{5}-1}$$

$$\Rightarrow \frac{c}{a} = \frac{3+\sqrt{5}}{2}$$

Q. **In ΔABC, $a^2(\cos^2 B - \cos^2 C) + b^2(\cos^2 C - \cos^2 A) + c^2(\cos^2 A - \cos^2 B)$ is equal to ?**

Sol. $a^2(\sin^2 C - \sin^2 B) + b^2(\sin^2 A - \sin^2 C) + c^2(\sin^2 B - \sin^2 A)$

$= [(a\sin c)^2 - (c\sin A)^2] + [(b\sin A)^2 - (a\sin B)^2] + [(c\sin B)^2 - (b \sin C)^2]$

$= 0 + 0 + 0 = 0$ $\quad \left\{\begin{matrix} \text{by sine rule} & a\sin B = b\sin A \\ & a\sin C = c\sin A \\ & b\sin C = c\sin B \end{matrix}\right\}$

Q. **If a^2, b^2, c^2 are in AP then tanA, tanB, tanC are in**

(A) H.P. **(B) G.P.** **(C) A.P.** **(D) None**

Ans. **(A)**

Sol. let $a^2 = b^2 - k$, $c^2 = b^2 + k$

$$\tan A = \frac{\sin A}{\cos A} = \frac{a}{2R} \cdot \frac{2bc}{\left(b^2 + c^2 - a^2\right)} = \frac{abc}{R\left(b^2 + 2k\right)}$$

$$\tan B = \frac{\sin B}{\cos B} = \frac{b}{2R} \cdot \frac{2ac}{a^2 + c^2 - b^2} = \frac{abc}{R\left(b^2\right)}$$

$$\tan C = \frac{\sin C}{\cos C} = \frac{c}{2R}\cdot\frac{2ab}{a^2+b^2-c^2} = \frac{abc}{R\left(b^2-2k\right)}$$

clearly, $\frac{1}{\tan A}, \frac{1}{\tan B}, \frac{1}{\tan C}$ are in AP

$\Rightarrow$ tanA, tanB, tanC in HP.

Q. **In a ΔABC, $\angle A = 60°$ and $b : c = \left(\sqrt{3}+1\right) : 2$ then find the value of $\angle B - \angle C$.**

Sol. $$\tan\left(\frac{B-C}{2}\right) = \frac{b-c}{b+c}\cot\frac{A}{2}$$

$$\Rightarrow \tan\left(\frac{\theta}{2}\right) = \frac{\left(\sqrt{3}+1\right)k-2k}{\left(\sqrt{3}+1\right)k+2k}\ \cot 30° \qquad (\angle B - \angle C = \theta)$$

$$\Rightarrow \tan\left(\frac{\theta}{2}\right) = \frac{\sqrt{3}-1}{\sqrt{3}\left(\sqrt{3}+1\right)}\cdot\sqrt{3} = \frac{\sqrt{3}-1}{\sqrt{3}+1} = \tan 15°$$

$$\Rightarrow \frac{\theta}{2} = 15° \Rightarrow \theta = 30°$$

$\angle B - \angle C = 30°$

Q. **If the sides of a triangle are p, q and $\sqrt{p^2+pq+q^2}$, then the biggest angle is:**

Sol. Let $r = \sqrt{p^2+pq+q^2}$

Opposite angle to r will be biggest, let it be "θ", then

$$\cos\theta = \frac{p^2+q^2-r^2}{2pq} = \frac{p^2+q^2-\left(p^2+pq+q^2\right)}{2pq}$$

$$\cos\theta = \frac{-1}{2} \Rightarrow \theta = \frac{2\pi}{3}$$

Q. **In a ΔABC, $\frac{\cos A}{a} = \frac{\cos B}{b} = \frac{\cos C}{c}$ and the side a = 2 then area of the triangle is:**

Sol. Using sine rule,

$$\frac{\cos A}{2R\sin A} = \frac{\cos B}{2R\sin B} = \frac{\cos C}{2R\sin C}$$

$\Rightarrow \cot A = \cot B = \cot C$

$\Rightarrow A = B = C$

Hence ΔABC is equilateral and $a = b = c = 2$, $A = B = C = \frac{\pi}{3}$

Area of the $\Delta = \frac{1}{2}$ ab sinC $= \frac{1}{2}(2\cdot 2)\sin\frac{\pi}{3}$

$= \sqrt{3}$ sq. units

Q. **In any triangle if $\frac{a^2 - b^2}{a^2 + b^2} = \frac{\sin(A-B)}{\sin(A+B)}$ then**

(A) The triangle is right angled
(B) The triangle is isosceles
(C) The triangle is equilateral
(D) Both (A) or (B) can be true

Ans. **(D)**

Sol. Using componendo and dividendo,

$$\frac{a^2 - b^2 + a^2 + b^2}{a^2 - b^2 - a^2 - b^2} = \frac{\sin(A-B) + \sin(A+B)}{\sin(A-B) - \sin(A+B)}$$

$$\Rightarrow \frac{-a^2}{b^2} = \frac{2\sin A\cos B}{-2\cos A\sin B}$$

$$\Rightarrow \frac{\sin^2 A}{\sin^2 B} = \frac{\sin A\cos B}{\cos A\sin B}$$

$$\Rightarrow \frac{\sin A}{\sin B} = \frac{\cos B}{\cos A}$$

$\Rightarrow \sin A\cos A - \sin B\cos B = 0$

$\Rightarrow \frac{1}{2}[\sin 2A - \sin 2B] = 0$

$\Rightarrow \cos(A + B)\sin(A - B) = 0$

$\cos(A + B) = 0$ or $\sin(A - B) = 0$

$A + B = \frac{\pi}{2}$ $\qquad$ $A - B = 0$

$C = \frac{\pi}{2}$ $\qquad$ $A = B$

Right angled Δ $\qquad$ Isosceles Δ

Q. **For any triangle ABC. If B = 3C, show that** $\cos C = \sqrt{\frac{b+c}{4c}}$ & $\sin\frac{A}{2} = \frac{b-c}{2c}$

Sol. $A + B + C = \pi \Rightarrow A + 4C = \pi \Rightarrow A = \pi - 4C$

$$\tan\left(\frac{B-C}{2}\right) = \frac{b-c}{b+c}\cot\frac{A}{2}$$

$$\tan\left(\frac{3C-C}{2}\right) = \frac{b-c}{b+c}\cot\left(\frac{\pi}{2} - 2C\right)$$

$$\tan C = \frac{b-c}{b+c}\tan 2C$$

$$\Rightarrow \tan C = \frac{b-c}{b+c}\cdot\frac{2\tan C}{1-\tan^2 C}$$

$$1 - \tan^2 C = \frac{2(b-c)}{b+c}$$

$$\Rightarrow 1 - (\sec^2 C - 1) = \frac{2(b-c)}{b+c}$$

$$\sec^2 C = 2 \times \frac{2c}{b+c} \Rightarrow \cos C = \sqrt{\frac{b+c}{4c}}$$

$$\text{Now } \sin\frac{A}{2} = \sin\left(\frac{\pi}{2} - 2C\right) = \cos 2C = 2\cos^2 C - 1$$

$$= 2\left(\frac{b+c}{4c}\right) - 1 = \frac{b-c}{2c}$$

Q. **ABCD is a trapezium such that AB, DC are parallel & BC is perpendicular to them. If angle ADB = θ, BC = p & CD = q then the value of AB is**

(A) $\frac{(p^2+q^2)\sin\theta}{p\cos\theta + q\sin\theta}$ **(B)** $\frac{(p^2+q^2)\sin\theta}{p\cos\theta - q\sin\theta}$ **(C)** $\frac{(p^2-q^2)\sin\theta}{p\cos\theta + q\sin\theta}$ **(D)** $\frac{2(p^2+q^2)\sin\theta}{p\cos\theta - q\sin\theta}$

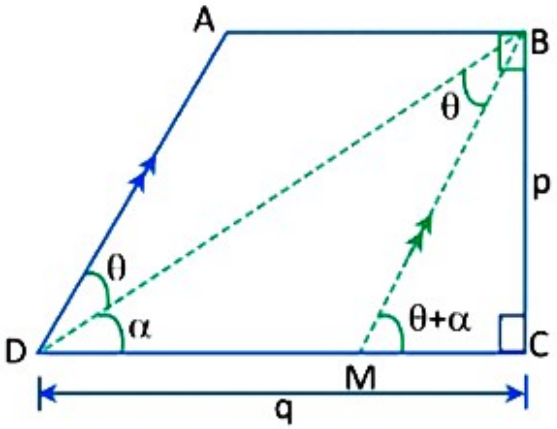
Ans. **(A)**

Sol. See the figure let BM || AD and $\angle BDM = \alpha$

now in ΔBDC

$\tan\alpha = p/q$ ...(1)

In ΔBMC



$$\tan(\theta + \alpha) = \frac{p}{MC}$$

$$\tan(\theta + \alpha) = \frac{p}{q - DM}$$

$$\frac{\tan\theta + \tan\alpha}{1 - \tan\theta\tan\alpha} = \frac{p}{q - AB}$$

$$\Rightarrow \frac{\tan\theta + \frac{p}{q}}{1 - \tan\theta\frac{p}{q}} = \frac{p}{q - AB}$$

$$\Rightarrow \frac{q\tan\theta + p}{q - p\tan\theta} = \frac{p}{q - AB}$$

$\Rightarrow q^2\tan\theta + pq - AB(q\tan\theta + p) = pq - p^2\tan\theta$

$\Rightarrow (q^2 + p^2)\tan\theta = (q\tan\theta + p)AB$

$$\Rightarrow AB = \frac{\left(p^2 + q^2\right)\sin\theta}{q\sin\theta + p\cos\theta}$$

Q. **If in a ΔABC, a = 6, b =3 and $\cos(A - B) = \frac{4}{5}$ then find its area ?**

Sol. Let $A - B = \theta$ then $\cos\theta = \frac{4}{5}$

$$\Rightarrow \frac{1 - \tan^2\frac{\theta}{2}}{1 + \tan^2\frac{\theta}{2}} = \frac{4}{5} \Rightarrow \tan\frac{\theta}{2} = \frac{1}{3}$$

$$\Rightarrow \tan\left(\frac{A - B}{2}\right) = \frac{a - b}{a + b}\cot\frac{C}{2}$$

$$\Rightarrow \frac{1}{3} = \frac{3}{9}\cot\frac{C}{2} \Rightarrow \cot\frac{C}{2} = 1 \Rightarrow C = 90°$$

Area of $\Delta = \frac{1}{2}ab\sin C = \frac{1}{2} \times 6 \times 3 \times 1 = 9$ sq. units

Q. **The sides of a triangle are consecutive integers n, n + 1 and n + 2 and the largest angle is twice the smallest angle. Find n?**

Sol. Let $a = n$, $b = n + 1$, $c = n + 2$

also given $C = 2A$ then $B = \pi - 3A$

now $\dfrac{\sin A}{a} = \dfrac{\sin B}{b} = \dfrac{\sin C}{c}$

$$\Rightarrow \quad \frac{\sin A}{n} = \frac{\sin(3A)}{n+1} = \frac{\sin 2A}{n+2}$$

$\Rightarrow \sin A = n\alpha, \sin 2A = (n + 2)\alpha, \sin 3A = (n + 1)\alpha$

Now $\sin 2A = 2\sin A \cos A$

$$(n + 2)\alpha = 2(n\alpha)\cos A \Rightarrow \cos A = \frac{n+2}{2n}$$

and $3\sin A - 4\sin^3 A = \sin 3A$

$\Rightarrow 3\sin A - 4\sin^3 A = (n + 1)\alpha$

$\Rightarrow \sin A\{3 - 4\sin^2 A\} = (n + 1)\alpha$

$$\Rightarrow \quad n\alpha\left\{3-4\left(1-\left(\frac{n+2}{2n}\right)^2\right)\right\} = (n+1)\alpha$$

$$\Rightarrow \quad n\left\{\left(\frac{n+2}{n}\right)^2 - 1\right\} = (n+1)$$

$\Rightarrow n\{(n + 2)^2 - n^2\} = n^2(n + 1)$

$\Rightarrow n^2 - 3n - 4 = 0$

$\Rightarrow (n - 4)(n + 1) = 0 \Rightarrow n = 4$

Q. **If a = 5, b = 7, c = 8 find angle B ?**

Sol. $$\cos B = \frac{5^2 + 8^2 - 7^2}{2(5)(8)}$$

$$= \frac{25 + 64 - 49}{80} = \frac{1}{2}$$

$$\Rightarrow \quad B = \frac{\pi}{3}$$

Q. **Prove that** $\frac{\cos A}{a}+\frac{\cos B}{b}+\frac{\cos C}{c}=\frac{a^2+b^2+c^2}{2abc}$

Sol. LHS

$$=\frac{b^2+c^2-a^2}{2abc}+\frac{a^2+c^2-b^2}{2abc}+\frac{a^2+b^2-c^2}{2abc}$$

$$=\frac{a^2+b^2+c^2}{2abc}=\text{RHS}$$

Hence Proved.

Q. **In a ΔABC, if the sides are a = 3, b = 5 and c = 4 then $\sin\frac{B}{2}+\cos\frac{B}{2}$ is equal to ?**

Sol. $\cos B=\frac{a^2+c^2-b^2}{2ac}=\frac{9+16-25}{2(3)(4)}=0$

$\Rightarrow \quad B=\frac{\pi}{2}$

hence $\sin\frac{B}{2}+\cos\frac{B}{2}=\sin\frac{\pi}{4}+\cos\frac{\pi}{4}$

$$=\frac{1}{\sqrt{2}}+\frac{1}{\sqrt{2}}=\sqrt{2}$$

Q. **If the radius of the circumcircle of an isosceles triangle PQR is equal to PQ (= PR) then the angle P is:**

Sol. Let PQ = PR = a

Now,

$2R'=\frac{PQ}{\sin R}=\frac{PR}{\sin Q}=\frac{QR}{\sin P}$, R′ = Radius of Circumcircle

$2a=\frac{a}{\sin R}=\frac{a}{\sin Q}=\frac{QR}{\sin P}$

$\sin R=\sin Q=\frac{1}{2}\Rightarrow \angle R=\angle Q=\frac{\pi}{6}$

then $\angle P=\pi-\left(\frac{\pi}{6}+\frac{\pi}{6}\right)=\frac{2\pi}{3}$

Q. **In a ΔABC, the value of $(b^2 - c^2)\cot A + (c^2 - a^2)\cot B + (a^2 - b^2)\cot C$ is**
(A) 0 (B) 1 (C) 2 (D) None of these

Ans. **(A)**

Sol. LHS is

I^{st} term $= \{(2R\sin B)^2 - (2R\sin C)^2\}\cot A$

$= 4R^2(\sin^2 B - \sin^2 C)\cot A$

$= 4R^2\{\sin(B + C)\sin(B - C)\}\cot A$

$= 4R^2[\sin A.\sin(B - C)]\dfrac{\cos A}{\sin A}$

$= 4R^2\{\sin(B - C)\cos(\pi - (B + C))\}$

$= -4R^2\{\sin(B - C)\cos(B + C)\}$

$= -2R^2\{\sin 2B - \sin 2C\}$

Similarly,

IInd term $= -2R^2\{\sin 2C - \sin 2A\}$

III term $= -2R^2\{\sin 2A - \sin 2B\}$

Sum of all terms $= 0$

Q. **Prove that $\cot A + \cot B + \cot C = \dfrac{a^2 + b^2 + c^2}{4\Delta}$**

Sol. LHS $\cot A + \cot B + \cot C$

$= \dfrac{\cos A}{\sin A} + \dfrac{\cos B}{\sin B} + \dfrac{\cos C}{\sin C}$

$= \dfrac{b^2 + c^2 - a^2}{2bc\sin A} + \dfrac{a^2 + c^2 - b^2}{2ac\sin B} + \dfrac{a^2 + b^2 - c^2}{2ab\sin C}$

$= \dfrac{b^2 + c^2 - a^2}{4\Delta} + \dfrac{a^2 + c^2 - b^2}{4\Delta} + \dfrac{a^2 + b^2 - c^2}{4\Delta}$

$= \dfrac{a^2 + b^2 + c^2}{4\Delta} = RHS$ **Hence Proved.**

Q. Let ABC be a triangle such that $\angle ACB = \frac{\pi}{6}$ and let a, b and c denote the length of sides opposite to A, B and C respectively. The value(s) of x for which $a = x^2 + x + 1$, $b = x^2 - 1$ and $c = 2x + 1$ is (are)

(A) $-\left(2+\sqrt{3}\right)$ **(B)** $1+\sqrt{3}$ **(C)** $2+\sqrt{3}$ **(D)** $4\sqrt{3}$

Ans. **(B)**

Sol.

$$\cos C = \frac{a^2+b^2-c^2}{2ab}$$

$$\Rightarrow \cos\frac{\pi}{6} = \frac{\left(x^2+x+1\right)^2+\left(x^2-1\right)^2-(2x+1)^2}{2\left(x^2+x+1\right)\left(x^2-1\right)}$$

$$\Rightarrow \frac{\sqrt{3}}{2} = \frac{\left(x^2+x+1\right)^2-(2x+1)^2+\left(x^2-1\right)^2}{2\left(x^2+x+1\right)\left(x^2-1\right)}$$

$$\Rightarrow \frac{\sqrt{3}}{2} = \frac{\left(x^2+3x+2\right)\left(x^2-x\right)+\left(x^2-1\right)^2}{2\left(x^2+x+1\right)\left(x^2-1\right)}$$

$$\Rightarrow \frac{\sqrt{3}}{2} = \frac{(x+1)(x+2)x(x-1)+(x+1)^2(x-1)^2}{2\left(x^2+x+1\right)(x+1)(x-1)}$$

$$\Rightarrow \frac{\sqrt{3}}{2} = \frac{x(x+2)+\left(x^2-1\right)}{2\left(x^2+x+1\right)}$$

$$\Rightarrow \sqrt{3}\left(x^2+x+1\right) = 2x^2+2x-1$$

$$\Rightarrow \left(2-\sqrt{3}\right)x^2+\left(2-\sqrt{3}\right)x-\left(1+\sqrt{3}\right)=0$$

$$x = \frac{-\left(2-\sqrt{3}\right)\pm\sqrt{\left(2-\sqrt{3}\right)^2+4\left(2-\sqrt{3}\right)\left(1+\sqrt{3}\right)}}{2\left(2-\sqrt{3}\right)}$$

$$x = \frac{-\left(2-\sqrt{3}\right)\pm\sqrt{\left(2-\sqrt{3}\right)\left(2-\sqrt{3}+4+4\sqrt{3}\right)}}{2\left(2-\sqrt{3}\right)}$$

$$x = \frac{-\left(2-\sqrt{3}\right) \pm \sqrt{\left(2-\sqrt{3}\right)3\left(2+\sqrt{3}\right)}}{2\left(2-\sqrt{3}\right)}$$

$$x = \frac{-\left(2-\sqrt{3}\right) \pm \sqrt{3}}{2\left(2-\sqrt{3}\right)} = \frac{2\left(\sqrt{3}-1\right)}{2\left(2-\sqrt{3}\right)}, \frac{-2}{2\left(2-\sqrt{3}\right)}$$

$$x = \left(\sqrt{3}-1\right)\left(2+\sqrt{3}\right), -\left(2+\sqrt{3}\right)$$

$$x = 1+\sqrt{3},$$

$x = -(2+\sqrt{3})$: rejected ($\because 2x + 1 < 0$)

Q. **If $a = \sqrt{3}, b = \frac{1}{2}\left(\sqrt{6}+\sqrt{2}\right)$ and $c = \sqrt{2}$ then find $\angle A$**

Sol.

$$\cos A = \frac{b^2+c^2-a^2}{2bc}$$

$$= \frac{\frac{1}{4}\left(8+4\sqrt{3}\right)+2-3}{\sqrt{2}\left(\sqrt{3}+1\right)\left(\sqrt{2}\right)}$$

$$= \frac{\left(2+\sqrt{3}\right)-1}{2\left(\sqrt{3}+1\right)} = \frac{1}{2} \Rightarrow A = \frac{\pi}{3}$$

Q. **The sides of a triangle are $x^2 + x + 1$, $2x + 1$ and $x^2 - 1$. Prove that the greatest angle is 120°**

Sol. Let $x^2 + x + 1 = a$, $2x + 1 = b$, $x^2 - 1 = c$

(i) $a > 0 \Rightarrow x^2 + x + 1 > 0$
$\Rightarrow x \in R$

(ii) $b > 0 \Rightarrow 2x + 1 > 0$
$\Rightarrow x > -\frac{1}{2}$

(iii) $c > 0 \Rightarrow x^2 - 1 > 0$
$\Rightarrow x \in (-\infty, -1) \cup (1, \infty)$

$\cap \Rightarrow x \in (1, \infty)$

Now for $x > 1$, side $x^2 + x + 1$ is the greatest side.
now

$$\cos A = \frac{b^2+c^2-a^2}{2bc}$$

$$= \frac{(2x+1)^2 + (x^2-1)^2 - (x^2+x+1)^2}{2(2x+1)(x^2-1)}$$

$$= \frac{(x^2+3x+2)(-x^2+x) + (x^2-1)^2}{2(2x+1)(x^2-1)}$$

$$= \frac{(x+1)(x+2)(-x)(x-1) + (x+1)^2(x-1)^2}{2(2x+1)(x+1)(x-1)}$$

$$= \frac{(-x^2-2x) + (x^2-1)}{2(2x+1)} = -\frac{1}{2} \Rightarrow A = 120°$$

Hence Proved.

Q. **In a triangle ABC if $a^2 + b^2 = 101\,c^2$ then find the value of $\frac{\cot C}{\cot A + \cot B}$.**

Sol.

$$\frac{\cot C}{\cot A + \cot B} = \frac{\frac{\cos C}{\sin C}}{\frac{\cos A}{\sin A} + \frac{\cos B}{\sin B}}$$

$$= \frac{\frac{a^2+b^2-c^2}{4\Delta}}{\frac{b^2+c^2-a^2}{4\Delta} + \frac{a^2+c^2-b^2}{4\Delta}} = \frac{a^2+b^2-c^2}{2c^2}$$

$$= \frac{101c^2 - c^2}{2c^2} = 50$$

Q. **The two adjacent sides of a cyclic quadrilateral are 2 & 5 and the angle between them is 60°. If the area of the quadrilateral is $4\sqrt{3}$, find remaining two sides.**

Sol. Let AD = 5, DC = 2
and $\angle D = 60°$
then $\angle B = 180° - 60°$
$= 120°$
Also let BC = x, AB = y
ar (ABCD) = ar(ΔADC) + ar(ΔABC)

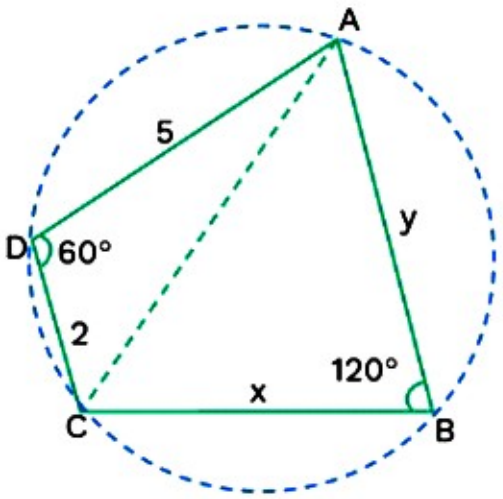
$$4\sqrt{3} = \frac{1}{2}\,2\cdot5\,\sin60° + \frac{1}{2}xy\,\sin120°$$

$$4\sqrt{3} = 5\left(\frac{\sqrt{3}}{2}\right) + \frac{1}{2}xy\left(\frac{\sqrt{3}}{2}\right)$$

$$\Rightarrow 16\sqrt{3} = 10\sqrt{3} + xy\sqrt{3}$$

$$\Rightarrow \quad xy + 10 = 16$$

$$xy = 6 \qquad \ldots(1)$$



In ΔADC

$$\cos60° = \frac{5^2 + 2^2 - AC^2}{2\times5\times2} \Rightarrow \frac{1}{2} = \frac{29 - AC^2}{20}$$

$$\Rightarrow AC^2 = 19$$

In ΔABC

$$\cos120° = \frac{x^2 + y^2 - AC^2}{2xy} \Rightarrow \frac{-1}{2} = \frac{x^2 + y^2 - 19}{2(6)}$$

$$x^2 + y^2 = 13 \qquad \ldots(2)$$

from (1) & (2) $\quad x = 3, y = 2$ or $x = 2, y = 3$

hence other sides are 2, 3.

Q. **Prove that $a(\cos B + \cos C) = 2(b + c)\sin^2\frac{A}{2}$**

Sol.

$$\frac{b+c}{a} = \frac{2R\sin B + 2R\sin C}{2R\sin A}$$

$$= \frac{2\sin\left(\frac{B+C}{2}\right)\cos\left(\frac{B-C}{2}\right)}{\sin A}$$

$$= \frac{2\cos\frac{A}{2}\cos\left(\frac{B-C}{2}\right)}{2\sin\frac{A}{2}\cos\frac{A}{2}} = \frac{2\cos\left(\frac{B-C}{2}\right)}{2\sin\frac{A}{2}}$$

$$= \frac{2\sin\left(\frac{A}{2}\right).\cos\left(\frac{B-C}{2}\right)}{2\sin^2\frac{A}{2}} = \frac{2\cos.\left(\frac{B+C}{2}\right).\cos\left(\frac{B-C}{2}\right)}{2\sin^2\frac{A}{2}}$$

$$= \frac{\cos B + \cos C}{2\sin^2\frac{A}{2}}$$

$$\Rightarrow a(\cos B + \cos C) = 2(b + c)\sin^2\frac{A}{2}$$

Hence Proved.

Q. **Prove that a(cosC − cosB) = 2(b − c) $\cos^2\frac{A}{2}$**

Sol.

$$\frac{b-c}{a}=\frac{2R\sin B-2R\sin C}{2R\sin A}$$

$$=\frac{2\cos\left(\frac{B+C}{2}\right).\sin\left(\frac{B-C}{2}\right)}{\sin A}$$

$$=\frac{2\sin\frac{A}{2}\sin\left(\frac{B-C}{2}\right)}{2\sin\frac{A}{2}\cos\frac{A}{2}}=\frac{2\sin\left(\frac{B-C}{2}\right)}{2\cos\frac{A}{2}}$$

$$=\frac{2\cos\left(\frac{A}{2}\right)\sin\left(\frac{B-C}{2}\right)}{2\cos^2\frac{A}{2}}=\frac{2\sin\left(\frac{B+C}{2}\right)\sin\left(\frac{B-C}{2}\right)}{2\cos^2\frac{A}{2}}$$

$$=\frac{\cos C-\cos B}{2\cos^2\frac{A}{2}}$$

$\Rightarrow a(\cos C-\cos B)=2(b-c)\cos^2\frac{A}{2}$ **Hence Proved.**

Q. **Prove that (b + c) cosA + (c + a) cosB + (a + b) cosC = a + b + c**

Sol.

LHS:

= (bcosA + acosB) + (bcosC + ccosB) + (ccosA + acosC)

= c + a + b **Hence Proved.**

Q. **Prove that 4R sinA sinB sinC = acosA + bcosB + ccosC**

Sol.

RHS

= 2RsinAcosA + 2RsinB cosB + 2RsinC cosC

= R(sin2A + sin2B + sin2C)

= R(4 sinA sinB sinC) **Hence Proved.**

Q. **If two times the square of the diameter of the circumcircle of a triangle is equal to the sum of the squares of its sides then prove that the triangle is right angled.**

Sol. Given

$2(2R)^2 = a^2 + b^2 + c^2$

$\Rightarrow 8R^2 = a^2 + b^2 + c^2$

$\Rightarrow 8R^2 = (2R\sin A)^2 + (2R\sin B)^2 + (2R\sin C)^2$

$\Rightarrow 8R^2 = 4R^2\{\sin^2 A + \sin^2 B + \sin^2 C\}$

$$\Rightarrow 2 = \frac{1-\cos 2A}{2} + \frac{1-\cos 2B}{2} + \frac{1-\cos 2C}{2}$$

$\Rightarrow \cos 2A + \cos 2B + \cos 2C = -1$

$\Rightarrow -1 - 4\cos A.\cos B.\cos C = -1$

$\Rightarrow \cos A = 0$ or $\cos B = 0$ or $\cos C = 0$

$$A = \frac{\pi}{2} \text{ or } B = \frac{\pi}{2} \text{ or } C = \frac{\pi}{2}$$

Hence triangle is right angled.

Q. **Prove that $a^3\cos(B - C) + b^3\cos(C - A) + c^3\cos(A - B) = 3abc$**

Sol. LHS

Ist term $= a^2[2R\sin A.\cos(B-C)]$

$= a^2R[2\sin(B + C)\cos(B - C)]$

$= a^2R[\sin 2B + \sin 2C]$

$= a^2R[2\sin B\cos B + 2\sin C\cos C]$

$= a^2[b\cos B + c\cos C]$

$= a^2b\cos B + a^2c\cos C$

Similarly

IInd term $= b^2c\cos C + b^2a\cos A$

IIIrd term $= c^2a\cos A + c^2b\cos B$

On adding all the three terms,

LHS $= ab(a\cos B + b\cos A) + bc(b\cos C + c\cos B) + ca(a\cos C + c\cos A)$

Now by projection Rule,

LHS $= ab(c) + bc(a) + ca(b) = 3abc$ **Hence Proved.**

HALF angle Formulas

1. (i) $\sin\left(\frac{A}{2}\right) = \sqrt{\frac{(s-b)(s-c)}{bc}}$

Proof: As we know that,

$$\cos A = \frac{b^2 + c^2 - a^2}{2bc}$$

$$\Rightarrow \quad 1 - 2\sin^2\frac{A}{2} = \frac{b^2 + c^2 - a^2}{2bc}$$

$$\Rightarrow \quad 2bc - 4bc\sin^2\frac{A}{2} = b^2 + c^2 - a^2$$

$$\Rightarrow \quad a^2 - (b^2 + c^2 - 2bc) = 4bc\sin^2\frac{A}{2}$$

$$\Rightarrow \quad a^2 - (b - c)^2 = 4bc\sin^2\frac{A}{2}$$

$$\Rightarrow \quad (a + b - c)(a - b + c) = 4bc\sin^2\frac{A}{2}$$

$$\Rightarrow \quad \sin^2\frac{A}{2} = \frac{\{(a+b+c) - 2c\}\{(a+b+c) - 2b\}}{4bc}$$

$$\Rightarrow \quad \sin^2\frac{A}{2} = \frac{(2s - 2c)(2s - 2b)}{4bc}$$

$$\Rightarrow \quad \sin^2\frac{A}{2} = \frac{(s-b)(s-c)}{bc}$$

$$\Rightarrow \quad \sin\frac{A}{2} = \sqrt{\frac{(s-b)(s-c)}{bc}} \qquad \left(\text{as } \sin\frac{A}{2} > 0\right)$$

(ii) $\sin\frac{B}{2} = \sqrt{\frac{(s-a)(s-c)}{ac}}$

(iii) $\sin\frac{C}{2} = \sqrt{\frac{(s-a)(s-b)}{ab}}$

2. (i) $\cos\frac{A}{2} = \sqrt{\frac{s(s-a)}{bc}}$ **(ii)** $\cos\frac{B}{2} = \sqrt{\frac{s(s-b)}{ac}}$

(iii) $\cos\frac{C}{2} = \sqrt{\frac{s(s-c)}{ab}}$

3. From the above formulae, we can prove:

(i) $\tan\frac{A}{2} = \sqrt{\frac{(s-b)(s-c)}{s(s-a)}}$ **(ii)** $\tan\frac{B}{2} = \sqrt{\frac{(s-a)(s-c)}{s(s-b)}}$

(iii) $\tan\frac{C}{2} = \sqrt{\frac{(s-a)(s-b)}{s(s-c)}}$

4. Area of triangle in terms of sides (Heron's formula)

$$\Delta = \frac{1}{2}bc\sin A = \frac{1}{2}bc\ 2\sin\frac{A}{2}\cdot\cos\frac{A}{2}$$

$$= bc\sqrt{\frac{(s-b)(s-c)}{bc}}\sqrt{\frac{s(s-a)}{bc}}$$

$$\therefore \quad \Delta = \sqrt{s(s-a)(s-b)(s-c)}$$

Note: Relation between area and circumradius $R = \frac{abc}{4\Delta}$

Proof: $\frac{abc}{4\Delta} = \frac{abc}{4\left(\frac{1}{2}ab\sin C\right)} = \frac{c}{2\sin C} = R$ **Hence Proved.**

Q. **Prove that** $\frac{abc}{s}\cos\frac{A}{2}\cdot\cos\frac{B}{2}\cdot\cos\frac{C}{2} = \Delta$

Sol. $$\text{LHS} = \frac{abc}{s}\sqrt{\frac{s(s-a)}{bc}}\sqrt{\frac{s(s-b)}{ac}}\sqrt{\frac{s(s-c)}{ab}}$$

$$= \sqrt{s(s-a)(s-b)(s-c)}$$

$$= \Delta$$ **Hence Proved.**

Q. **The circum-radius of the triangle whose sides are 13, 12 and 5 is:**

Sol. Let $a = 13, b = 12, c = 5$ then

as $a^2 = b^2 + c^2$ hence $\Rightarrow A = \frac{\pi}{2}$

$$\Delta = \frac{1}{2}\ bc\ \sin A = 30$$

$$\text{now } R = \frac{abc}{4\Delta} = \frac{13.12.5}{4\cdot 30} = \frac{13}{2} \text{ units}$$

Q. **If the cotangents of half the angles of a triangle are in A.P. then prove that the sides are in A.P.**

Sol. Given $\cot\frac{A}{2}, \cot\frac{B}{2}, \cot\frac{C}{2}$ in A.P.

$$\Rightarrow \quad 2\cot\frac{B}{2} = \cot\frac{A}{2} + \cot\frac{C}{2}$$

$$\Rightarrow \quad 2\sqrt{\frac{s(s-b)}{(s-a)(s-c)}} = \sqrt{\frac{s(s-a)}{(s-b)(s-c)}} + \sqrt{\frac{s(s-c)}{(s-a)(s-b)}}$$

Multiply by $\sqrt{s(s-a)(s-b)(s-c)}$

$\Rightarrow$ $2s(s - b) = s(s - a) + s(s - c)$
$\Rightarrow$ $2s^2 - 2sb = s^2 - sa + s^2 - sc$
$\Rightarrow$ $s(2b - a - c) = 0$
$\Rightarrow$ $2b = a + c \Rightarrow$ a, b, c in AP.

Q. **Prove that $b\cos^2\frac{C}{2} + c\cos^2\frac{B}{2} = s$**

Sol. LHS

$$= b\left\{\frac{s(s-c)}{ab}\right\} + c\left\{\frac{s(s-b)}{ac}\right\}$$

$$= \frac{1}{a}\left\{s(s-c) + s(s-b)\right\}$$

$$= \frac{1}{a}\{2s^2 - s(b + c)\}$$

$$= \frac{1}{a}\left\{2s^2 - s(2s-a)\right\} = \frac{as}{a} = s$$

Hence Proved.

Q. **If $c^2 = a^2 + b^2$ then prove that $4s(s - a)(s - b)(s - c) = a^2b^2$.**

Sol. $c^2 = a^2 + b^2 \Rightarrow c = \frac{\pi}{2}$

Now LHS is
$4\ s(s - a)(s - b)(s - c) = 4\Delta^2$

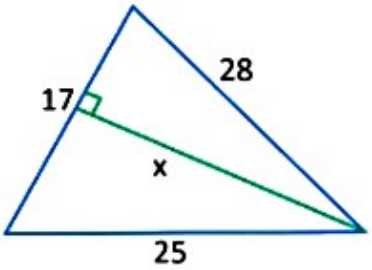
$$= 4\left(\frac{1}{2}ab\ \sin C\right)^2$$

$$= a^2b^2$$

Hence Proved.

Q. **If the sides of a triangle are 17, 25 and 28 then find the greatest length of altitude ?**

Sol. Greatest altitude will form on shortest side
Hence let greatest altitude is x, hence



$$\text{Area} = \frac{1}{2}.17.x = \Delta \qquad ...(1)$$

$$\text{Now } s = \frac{17+28+25}{2} = 35$$

$$\Delta = \sqrt{s(s-a)(s-b)(s-c)} = \sqrt{35.10.7.18}$$

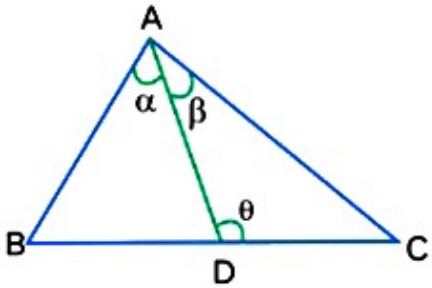
$$= 7.5.3.2 = 210 \text{ sq. unit} \qquad ...(2)$$

from (1) & (2)

$$\frac{1}{2}17x = 210 \Rightarrow x = \frac{420}{17} \text{ units}$$

'm – n' Theorem (Cotangent Theorem)

In ΔABC, D is a point on BC such that
BD : DC ≡ m : n then
$(m + n)\cot\theta = m\cot\alpha - n\cot\beta$
or $(m + n)\cot\theta = n\cot B - m\cot C$



m : n

Proof: In ΔABD $\quad \frac{BD}{\sin\alpha} = \frac{AD}{\sin B} \qquad ...(1)$

In ΔACD $\quad \frac{DC}{\sin} = \frac{AD}{\sin C} \qquad ...(2)$

(1) ÷ (2) $\Rightarrow$ $\quad \frac{BD}{DC}\frac{\sin\beta}{\sin\alpha} = \frac{\sin C}{\sin B} \qquad ...(3)$

$$\Rightarrow \quad \frac{m}{n}\frac{\sin\beta}{\sin\alpha} = \frac{\sin\{\pi-(\theta+\beta)\}}{\sin(\theta-\alpha)} \quad \{\because \alpha + B = \theta\}$$

$$\Rightarrow \quad \frac{m\sin\beta}{n\sin\alpha} = \frac{\sin(\theta+\beta)}{\sin(\theta-\alpha)}$$

$$\Rightarrow \quad \frac{m\sin\beta}{n\sin\alpha} = \frac{\sin\theta\cos\beta+\cos\theta\sin\beta}{\sin\theta\cos\alpha-\cos\theta\sin\alpha}$$

$$\Rightarrow \quad \frac{m\sin\beta}{n\sin\alpha} = \frac{\sin\theta\sin\beta(\cot\beta+\cot\theta)}{\sin\theta\sin\alpha(\cot\alpha-\cot\theta)}$$

$\Rightarrow$ $m\cot\alpha - m\cot\theta = n\cot\beta + n\cot\theta$

$\Rightarrow$ $(m+n)\cot\theta = m\cot\alpha - n\cot\beta$ **Hence Proved.**

from (3)

$$\Rightarrow \frac{m}{n}\frac{\sin(\pi-(\theta+C))}{\sin(\theta-B)} = \frac{\sin C}{\sin B}$$

$$\Rightarrow \frac{m(\sin\theta\cos C + \cos\theta\sin C)}{n(\sin\theta\cos B - \cos\theta\sin B)} = \frac{\sin C}{\sin B}$$

$$\Rightarrow \frac{m}{n}\frac{\sin\theta\sin C(\cot C + \cot\theta)}{\sin\theta\sin B(\cot C\cot\theta)} = \frac{\sin C}{\sin B}$$

$\Rightarrow$ $(m+n)\cot\theta = n\cot B - m\cot C$ **Hence Proved.**

Q. **Median from vertex C on opposite side is perpendicular to AC then prove that $2\tan A + \tan C = 0$**

Sol. $\frac{AD}{DB} = \frac{1}{1}$ ($\because$ CD is median)

$\Rightarrow m = 1, n = 1$

let $\angle CDB = \theta$

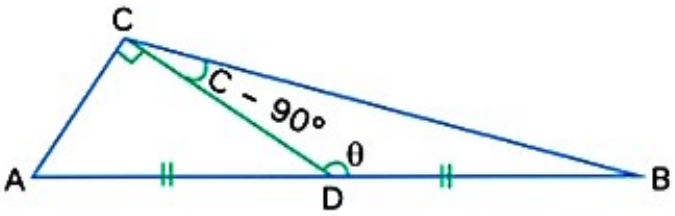
then by 'M – N' theorem

$(m+n)\cot\theta = m\cot\alpha - n\cot\beta$

$2\cot(90° + A) = 1.\cot 90° - 1.\cot(C - 90°)$

$-2\tan A = 0 + \tan C$

$\Rightarrow 2\tan A + \tan C = 0$



Q. **If the median AM of a ΔABC subtends an angle $\frac{3\pi}{4}$ with the side BC, then evaluate $|\cot B - \cot C|$**

Sol. Given AM is a median.

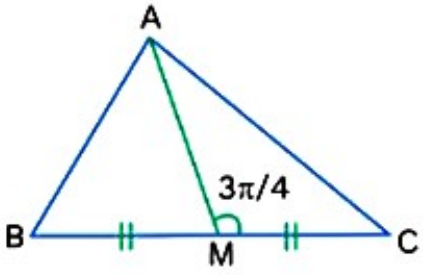
$\Rightarrow$ BM : MC = 1 : 1

$m = 1, n = 1$

and $\theta = \frac{3\pi}{4}$

by 'M – N' rule

$(m+n)\cot\theta = n\cot B - m\cot C$



$$2\cot\frac{3\pi}{4} = 1\cot B - 1\cot C$$

$$2(-1) = \cot B - \cot C$$

$$\Rightarrow |\cot B - \cot C| = |-2| = 2$$

Q. **The base of a triangle is divided into three equal parts. If t_1, t_2, t_3 are the tangents of the angles subtended by these parts at the opposite vertex then prove that**

$$\left(\frac{1}{t_1}+\frac{1}{t_2}\right)\left(\frac{1}{t_2}+\frac{1}{t_3}\right) = 4\left(1+\frac{1}{t_2^2}\right)$$

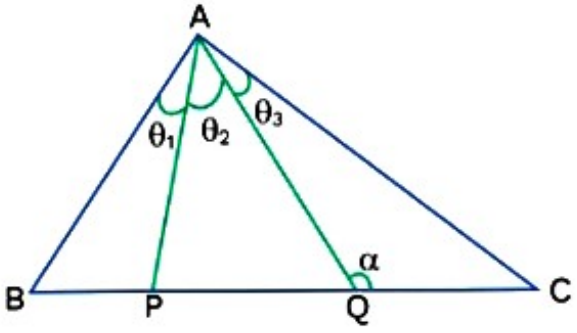
Sol. Given that

$\tan\theta_1 = t_1$

$\tan\theta_2 = t_2$

$\tan\theta_3 = t_3$

and BP : PQ : QC = 1 : 1 : 1



In ΔABC by 'm – n' Rule

$$(2+1)\cot\alpha = 2\cot(\theta_1+\theta_2) - 1\cot\theta_3$$

$$3\cot\alpha = 2\cot(\theta_1+\theta_2) - \cot\theta_3 \quad ...(1)$$

In ΔPAC by 'm – n' Rule

$$(1+1)\cot\alpha = 1\cot\theta_2 - 1\cot\theta_3$$

$$2\cot\alpha = \cot\theta_2 - \cot\theta_3 \quad ...(2)$$

(1) ÷ (2)
$$\frac{3}{2} = \frac{2\cot(\theta_1+\theta_2) - \cot\theta_3}{\cot\theta_2 - \cot\theta_3}$$

$$\Rightarrow 3\cot\theta_2 - 3\cot\theta_3 = 4\cot(\theta_1+\theta_2) - 2\cot\theta_3$$

$$\Rightarrow 3\cot\theta_2 - \cot\theta_3 = \frac{4(\cot\theta_2\cot\theta_1 - 1)}{\cot\theta_2 + \cot\theta_1}$$

$$\Rightarrow 3\cot^2\theta_2 + 3\cot\theta_1\cot\theta_2 - \cot\theta_3\cot\theta_2 - \cot\theta_1\cot\theta_3 = 4\cot\theta_2\cot\theta_1 - 4$$

$$\Rightarrow 3\cot^2\theta_2 + 4 = \cot\theta_1\cot\theta_2 + \cot\theta_2\cot\theta_3 + \cot\theta_1\cot\theta_3$$

$$\Rightarrow 4\cot^2\theta_2 + 4 = \cot\theta_1\cot\theta_2 + \cot^2\theta_2 + \cot\theta_1\cot\theta_3 + \cot\theta_2\cot\theta_3$$

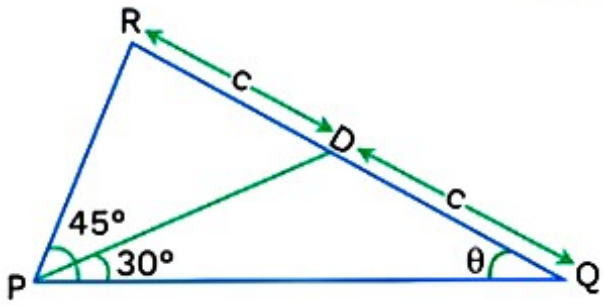
$$\Rightarrow 4(1+\cot^2\theta_2) = (\cot\theta_1+\cot\theta_2)(\cot\theta_2+\cot\theta_3)$$

$$\Rightarrow 4\left(1+\frac{1}{t_2^2}\right) = \left(\frac{1}{t_1}+\frac{1}{t_2}\right)\left(\frac{1}{t_2}+\frac{1}{t_3}\right)$$

Hence Proved.

Q. **A man observes that when he moves up a distance 'c' meters on a slope, then angle of depression of a point on the horizontal plane from the base of the slope is 30°, and when he moves up further a distance 'c' meters, the angle of depression of that point is 45°. The angle of inclination of the slope with the horizontal is**

Sol. Given



QD = DR = c meter

and $\angle DPQ = 30°$

$\angle RPQ = 45°$

now $\frac{DQ}{DR} = \frac{c}{c} = \frac{1}{1} \Rightarrow m = 1, n = 1$

and $\angle RDP = 30° + \theta$

now by 'M – N' Rule

$(m + n)\cot(30° + \theta) = m\cot 30° - n\cot(45° - 30°)$

$$\frac{2}{\tan(30° + \theta)} = 1.\sqrt{3} - 1.\left(2 + \sqrt{3}\right)$$

$\Rightarrow \tan(30° + \theta) = -1$

$\Rightarrow 30° + \theta = 135° \Rightarrow \theta = 105°$

Incircle of a triangle:

(i) $\Delta = sr \Rightarrow r = \frac{\Delta}{s}$

(ii) To express the radius of incircle in terms of sides and tangent of the half angle.

Hence

$$r = (s - a)\tan\frac{A}{2} = (s - b)\tan\frac{B}{2} = (s - c)\tan\frac{C}{2}$$

(iii) $$r = \frac{a\sin\left(\frac{B}{2}\right)\sin\left(\frac{C}{2}\right)}{\cos\left(\frac{A}{2}\right)}$$

(iv) $$r = 4R\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2}$$

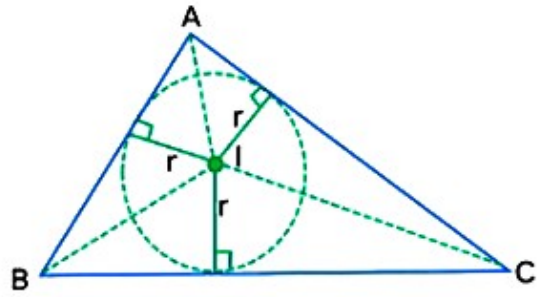
Proof:

(i) Area of triangle ABC

$\Delta = ar(AIB) + ar(BIC) + ar(AIC)$

$\Delta = \frac{1}{2}\, c.r + \frac{1}{2}\, a.r + \frac{1}{2}\, b.r$

$\Delta = \frac{1}{2}\,(a + b + c).r \Rightarrow \Delta = s.r \Rightarrow r = \frac{\Delta}{s}$



(ii) $(s - a)\tan\frac{A}{2} = (s - a)\sqrt{\frac{(s-b)(s-c)}{s(s-a)}}$

$= \frac{\sqrt{s(s-a)(s-b)(s-c)}}{s} = \frac{\Delta}{s} = r$

(iii) $\frac{a\sin\frac{B}{2}\sin\frac{C}{2}}{\cos\frac{A}{2}}$

$= a\sqrt{\frac{(s-a)(s-c)}{ac}}\cdot\sqrt{\frac{(s-a)(s-b)}{ab}}\sqrt{\frac{bc}{s(s-a)}} = \frac{\Delta}{s} = r$

(iv) $4R\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2}$

$= 4\frac{abc}{4\Delta}\sqrt{\frac{(s-b)(s-c)}{bc}}\sqrt{\frac{(s-a)(s-c)}{ac}}\sqrt{\frac{(s-a)(s-b)}{ab}}$

$= \frac{s(s-a)(s-b)(s-c)}{\Delta(s)} = \frac{\Delta^2}{\Delta s} = \frac{\Delta}{s} = r$

Radii Of The Escribed Circles:

(i) $r_1 = \frac{\Delta}{s-a},\ r_2 = \frac{\Delta}{s-b},\ r_3 = \frac{\Delta}{s-c}$

(ii) $r_1 = s\tan\left(\frac{A}{2}\right)$

$r_2 = s\tan\left(\frac{B}{2}\right)$

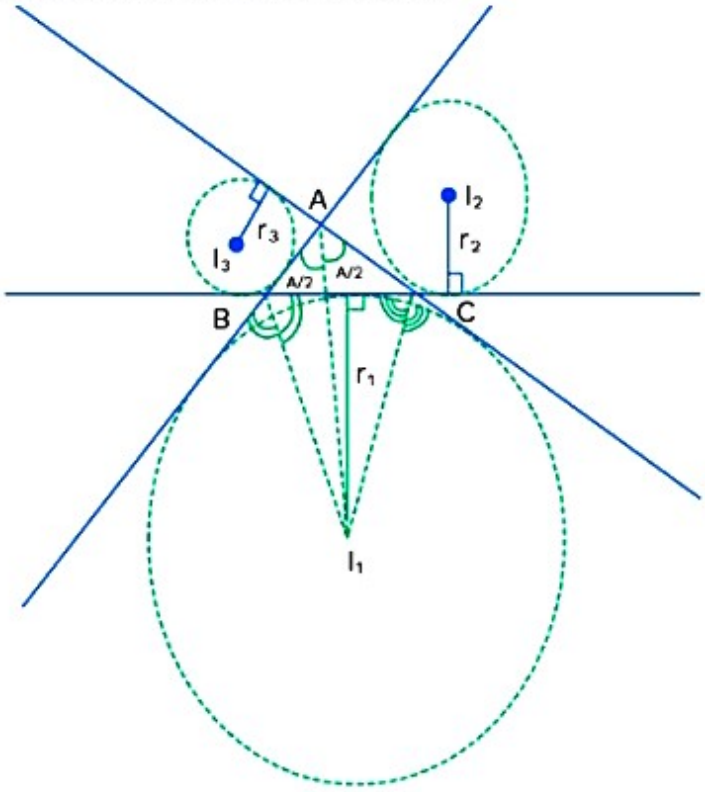
$r_3 = s\tan\left(\frac{C}{2}\right)$

(iii) Radii of the escribed circles in terms of one side and function of half angles:

$$r_1 = 4R\sin\left(\frac{A}{2}\right)\cos\left(\frac{B}{2}\right)\cos\left(\frac{C}{2}\right)$$

$$r_2 = 4R\cos\left(\frac{A}{2}\right)\sin\left(\frac{B}{2}\right)\cos\left(\frac{C}{2}\right)$$

$$r_3 = 4R\cos\left(\frac{A}{2}\right)\cos\left(\frac{B}{2}\right)\sin\left(\frac{C}{2}\right)$$



Proof: Area of triangle ABC

$$\Delta = ar(ABI_1) + ar(ACI_1) - ar(BCI_1)$$

$$\Delta = \frac{1}{2}c.r_1 + \frac{1}{2}b.r_1 - \frac{1}{2}a.r_1$$

$$\Delta = \frac{1}{2}(c + b - a)r_1$$

$$\Rightarrow \Delta = \frac{1}{2}(a + b + c - 2a)r_1$$

$$\Delta = \frac{1}{2}(2s - 2a)r_1 \Rightarrow \Delta = (s - a).\, r_1$$

$$\Rightarrow r_1 = \frac{\Delta}{s-a}$$

Hence other results can be proved by using angle formulae.

Q. **Prove that** $\frac{1}{r_1} + \frac{1}{r_2} + \frac{1}{r_3} = \frac{1}{r}$

Sol.

$$\text{LHS} = \frac{s-a}{\Delta} + \frac{s-b}{\Delta} + \frac{s-c}{\Delta}$$

$$= \frac{3s - (a+b+c)}{\Delta}$$

$$= \frac{3s - 2s}{\Delta}$$

$$= \frac{s}{\Delta} = \frac{1}{r}$$

Hence proved.

Q. **Prove that** $r_2r_3 + r_3r_1 + r_1r_2 = s^2$

Sol.

$$\text{LHS} = r_1r_2r_3\left(\frac{1}{r_1} + \frac{1}{r_2} + \frac{1}{r_3}\right)$$

$$= r_1r_2r_3 \cdot \frac{1}{r}$$

$$= \frac{\Delta^3}{(s-a)(s-b)(s-c)} \cdot \frac{s}{\Delta}$$

$$= \frac{\Delta^2}{s(s-a)(s-b)(s-c)} \cdot s^2 = s^2$$ **Hence proved.**

Q. **Prove that acotA + bcotB + ccotC = 2(R + r)**

Sol. $LHS = 2R\sin A \cdot \frac{\cos A}{\sin A} + 2R\sin B \cdot \frac{\cos B}{\sin B} + 2R\sin C \cdot \frac{\cos C}{\sin C}$

$= 2R(\cos A + \cos B + \cos C)$

$= 2R\left(1 + 4\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2}\right)$

$= 2R\left(1 + \frac{r}{R}\right) \quad \left(\text{as } r = 4R\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2}\right)$

$= 2(R + r)$ **Hence proved.**

Q. **Prove that** $(r + r_1)\tan\frac{B-C}{2} + (r + r_2)\tan\frac{C-A}{2} + (r + r_3)\tan\frac{A-B}{2} = 0$

Sol. In LHS

I^{st} term $= \left\{(s-a)\tan\frac{A}{2} + s\tan\frac{A}{2}\right\}\frac{b-c}{b+c}\cot\frac{A}{2}$

$= (s - a + s)\frac{b-c}{b+c}$

$= (2s - a).\frac{b-c}{b+c}$

$= (b + c).\frac{b-c}{b+c} = b - c$

similarly II term = c − a

III term = a − b

Hence sum of all terms = 0

Q. **Prove that** $\frac{bc - r_2r_3}{r_1} = \frac{ca - r_3r_1}{r_2} = \frac{ab - r_1r_2}{r_3} = r$

Sol. $rr_1 + r_2r_3 = \frac{\Delta^2}{s(s-a)} + \frac{\Delta^2}{(s-b)(s-c)}$

$= \Delta^2\left\{\frac{1}{s(s-a)} + \frac{1}{(s-b)(s-c)}\right\}$

$= \Delta^2\left\{\frac{(s-b)(s-c)+s(s-a)}{s(s-a)(s-b)(s-c)}\right\}$

$= \Delta^2\frac{\left(2s^2 - (a+b+c)s + bc\right)}{\Delta^2}$

$= = 2s^2 - (2s)s + bc$

$= bc$

$\Rightarrow rr_1 + r_2r_3 = bc$

$\Rightarrow rr_1 = bc - r_2r_3$

$\Rightarrow r = \frac{bc - r_2r_3}{r_1}$

Similarly, others can be proved.

Q. **Prove that** $rr_1r_2r_3 = \Delta^2 = r^2r_1^2\cot^2\frac{A}{2}$

Sol. $rr_1r_2r_3 = \frac{\Delta^4}{s(s-a)(s-b)(s-c)}$

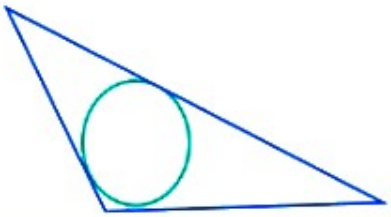
$= \frac{\Delta^4}{\Delta^2} = \Delta^2$

now $r^2r_1^2\cot^2\frac{A}{2} = \frac{\Delta^2}{s^2}\cdot\frac{\Delta^2}{(s-a)^2}\cdot\frac{1}{\tan^2\frac{A}{2}}$

$= \frac{\Delta^2}{s^2(s-a)^2} \times \frac{s(s-a)}{(s-b)(s-c)}$

$= \frac{\Delta^4}{s(s-a)(s-b)(s-c)} = \Delta^2$ **Hence proved.**

Q. **The triangle shown has sides of length 13, 30 and 37. The radius of the inscribed circle is**



(A) $7+\sqrt{2}$ **(B)** $\frac{9}{2}$

(C) $7-\sqrt{2}$ **(D)** $\frac{7}{2}$

Ans. **(B)**

Sol. Let a = 13, b = 30, c = 37

$$s = \frac{a+b+c}{2} = 40$$

$$\Delta = \sqrt{40.27.10.3} = 180$$

Inradius: $r = \frac{\Delta}{s} = \frac{180}{40} \Rightarrow r = \frac{9}{2}$ unit

Q. **In any triangle, prove that $\cos A + \cos B + \cos C = 1 + \frac{r}{R}$**

Sol. LHS

$$= 1 + 4\sin\frac{A}{2}\sin\frac{B}{2}.\sin\frac{C}{2}$$

$$= 1 + \frac{r}{R} \qquad \left(\text{as } r = 4R\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2}\right)$$

Hence Proved.

Q. **Prove that $\frac{r_1 + r_2}{1+\cos C} = 2R$**

Sol. LHS

$$= \frac{4R\cos\frac{C}{2}}{2\cos^2\frac{C}{2}}\left(\sin\frac{A}{2}\cos\frac{B}{2} + \cos\frac{A}{2}\sin\frac{B}{2}\right)$$

$$= \frac{2R}{\cos\frac{C}{2}}\sin\left(\frac{A+B}{2}\right)$$

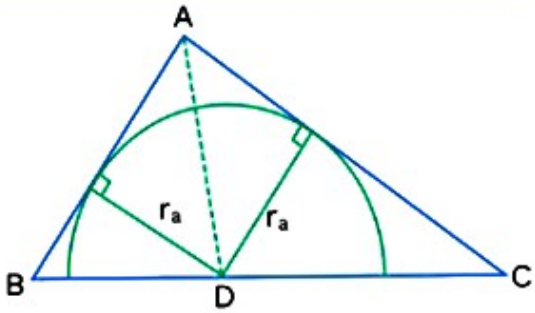
$$= \frac{2R}{\cos\frac{C}{2}} \cdot \sin\left(\frac{\pi - C}{2}\right) = 2R$$

Hence Proved.

Q. **In acute angled triangle ABC, a semicircle with radius r_a is constructed with its base on BC and tangent to the other two sides r_b and r_c are defined similarly. If r is the radius of the incircle of triangle ABC then prove that**

$$\frac{2}{r} = \frac{1}{r_a} + \frac{1}{r_b} + \frac{1}{r_c}$$

Sol.



$\Delta = ar(\Delta ABD) + ar(\Delta ACD)$

$$\Delta = \frac{1}{2} AB.r_a + \frac{1}{2} AC.r_a$$

$$\Delta = \frac{1}{2}(b + c)\, r_a$$

$$\Rightarrow r_a = \frac{2\Delta}{b+c}$$

Similarly, $r_b = \frac{2\Delta}{a+c}, r_c = \frac{2\Delta}{a+b}$

$$\frac{1}{r_a} + \frac{1}{r_b} + \frac{1}{r_c} = \frac{b+c}{2\Delta} + \frac{a+c}{2\Delta} + \frac{a+b}{2\Delta}$$

$$= \frac{1}{2\Delta} 2(a + b + c)$$

$$= \frac{2s}{\Delta} = \frac{2}{r}$$

Hence Proved.

Q. **Prove that** $\frac{r_1}{bc} + \frac{r_2}{ca} + \frac{r_3}{ab} = \frac{1}{r} - \frac{1}{2R}$

Sol.

In LHS

$$\text{I term} = \frac{4R\sin\frac{A}{2}\cos\frac{B}{2}\cos\frac{C}{2}}{2R\sin B.2R\sin C}$$

$$= \frac{1}{R} \frac{\sin\frac{A}{2}\cos\frac{B}{2}\cos\frac{C}{2}}{\left(2\sin\frac{B}{2}\cos\frac{B}{2}\right)\left(2\sin\frac{C}{2}\cos\frac{C}{2}\right)}$$

$$= \frac{1}{R} \frac{\sin\frac{A}{2}}{4\sin\frac{B}{2}\sin\frac{C}{2}}$$

Similarly,

$$\text{II term} = \frac{1}{R}\frac{\sin\frac{B}{2}}{4\sin\frac{A}{2}\sin\frac{C}{2}}$$

$$\text{III term} = \frac{1}{R}\frac{\sin\frac{C}{2}}{4\sin\frac{A}{2}\sin\frac{B}{2}}$$

Sum of all terms

$$= \frac{1}{R}\frac{\sin^2\frac{A}{2}+\sin^2\frac{B}{2}+\sin^2\frac{C}{2}}{4\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2}}$$

$$= \frac{1}{R}\frac{(1-\cos A)+(1-\cos B)+(1-\cos C)}{2.4\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2}}$$

$$= \frac{1}{R}\frac{3-\left(1+4\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2}\right)}{8\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2}}$$

$$= \frac{1}{R}\frac{\left(2-\frac{r}{R}\right)}{2\left(\frac{r}{R}\right)} = \frac{1}{r}-\frac{1}{2R}$$

Hence Proved.

Q. **Prove that $r_1 + r_2 + r_3 - r = 4R$**

Sol. LHS

$$= \left\{\frac{\Delta}{s-a}+\frac{\Delta}{s-b}\right\}+\left\{\frac{\Delta}{s-c}-\frac{\Delta}{s}\right\}$$

$$= \Delta\left\{\frac{2s-a-b}{(s-a)(s-b)}\right\}+\Delta\left\{\frac{s-(s-c)}{s(s-c)}\right\}$$

$$= \frac{\Delta c}{(s-a)(s-b)}+\frac{\Delta c}{s(s-c)}$$

$$= \Delta c\left\{\frac{2s^2-(a+b+c)s+ab}{s(s-a)(s-b)(s-c)}\right\}$$

$$= \frac{\Delta(abc)}{\Delta^2} = \frac{abc}{4\Delta}.4 = 4R$$ **Hence Proved.**

Q. **The area of a triangle is 96 sq. units and the measure of its ex radius are 8, 12, 24. Find the length of the sides.**

Sol. Given $\Delta = 96$, $r_1 = 8$, $r_2 = 12$, $r_3 = 24$

Now $r_1 = \frac{\Delta}{s-a} \Rightarrow s - a = \frac{96}{8} \Rightarrow s - a = 12$ ...(1)

$s - b = \frac{96}{12} \Rightarrow s - b = 8$ ...(2)

$s - c = \frac{96}{24} \Rightarrow s - c = 4$ ...(3)

Adding (1), (2) & (3)
$\Rightarrow 3s - (a + b + c) = 24$
Hence $3s - 2s = 24 \Rightarrow s = 24$
from (1), $a = 12$
from (2), $b = 16$
from (3), $c = 20$

Q. **If $\left(1-\frac{r_1}{r_2}\right)\left(1-\frac{r_1}{r_3}\right) = 2$, prove that the triangle is right angled.**

Sol. $\left(1-\frac{s-b}{s-a}\right)\left(1-\frac{s-c}{s-a}\right) = 2$

$$\Rightarrow \frac{(s-a)-(s-b)}{(s-a)}\cdot\frac{(s-a)-(s-c)}{(s-a)} = 2$$

$\Rightarrow (b - a)(c - a) = 2(s - a)^2$

$$\Rightarrow bc - (b + c)a + a^2 = 2\left\{\frac{(b+c)-a}{2}\right\}^2$$

$\Rightarrow 2bc - 2(b + c)a + 2a^2 = (b + c)^2 - 2(b + c)a + a^2$

$\Rightarrow a^2 = b^2 + c^2 \Rightarrow \angle A = \frac{\pi}{2}$

$\Rightarrow \Delta ABC$ is right angled.

Q. **Prove that $r^2 + r_1^2 + r_2^2 + r_3^2 = 16R^2 - (a^2 + b^2 + c^2)$**

Sol. We know that

$r_1 + r_2 + r_3 - r = 4R$

on squaring we get,

$r^2 + r_1^2 + r_2^2 + r_3^2 + 2(r_1r_2 + r_2r_3 + r_3r_1) - 2r(r_1 + r_2 + r_3) = 16R^2$

$\Rightarrow r^2 + r_1^2 + r_2^2 + r_3^2 + 2s^2 - 2(rr_1 + rr_2 + rr_3) = 16R^2$ ...(1)

now $rr_1 + rr_2 + rr_3 = \frac{\Delta^2}{s(s-a)} + \frac{\Delta^2}{s(s-b)} + \frac{\Delta^2}{s(s-c)}$

$= \Delta^2 \left\{ \frac{(s-b)(s-c) + (s-a)(s-c) + (s-a)(s-b)}{s(s-a)(s-b)(s-c)} \right\}$

$= 3s^2 - 2(a + b + c)s + (ab + bc + ca)$

$= -s^2 + (ab + bc + ca)$ ...(2)

put value from (2) into (1) then

$r^2 + r_1^2 + r_2^2 + r_3^2 + 2s^2 - 2(-s^2 + ab + bc + ca) = 16R^2$

$r^2 + r_1^2 + r_2^2 + r_3^2 + 4\left(\frac{a+b+c}{2}\right)^2 - 2(ab + bc + ca) = 16R^2$

$\Rightarrow r^2 + r_1^2 + r_2^2 + r_3^2 = 16R^2 - (a^2 + b^2 + c^2)$ **Hence Proved.**

Q. **Prove that $\frac{b-c}{r_1} + \frac{c-a}{r_2} + \frac{a-b}{r_3} = 0$**

Sol. In LHS

I term $= \frac{2R}{r_1}(\sin B - \sin C)$

$= \frac{2R\left(2\cos\left(\frac{B+C}{2}\right)\sin\left(\frac{B-C}{2}\right)\right)}{4R\sin\frac{A}{2}\cos\frac{B}{2}\cos\frac{C}{2}}$

$= \frac{\sin\frac{B}{2}\cos\frac{C}{2} - \cos\frac{B}{2}\sin\frac{C}{2}}{\cos\frac{B}{2}.\cos\frac{C}{2}}$

$$= \tan\frac{B}{2} - \tan\frac{C}{2}$$

Similarly II term $= \tan\frac{C}{2} - \tan\frac{A}{2}$

III term $= \tan\frac{A}{2} - \tan\frac{B}{2}$

$\Rightarrow$ Sum of all terms = 0

Q. **If Δ is the area of a triangle with side lengths a, b, c then show that $\Delta \le \frac{1}{4}\sqrt{(a+b+c)abc}$. Also show that equality occurs in the above inequality if and only if a = b = c**

Sol. By using AM $\ge$ GM

$$\frac{(s-a)+(s-b)}{2} \ge \sqrt{(s-a)(s-b)} \quad \ldots(1)$$

$$\frac{(s-b)+(s-c)}{2} \ge \sqrt{(s-b)(s-c)} \quad \ldots(2)$$

$$\frac{(s-c)+(s-a)}{2} \ge \sqrt{(s-c)(s-a)} \quad \ldots(3)$$

On multiplying (1), (2) and (3) we get

$$\frac{c}{2}.\frac{a}{2}.\frac{b}{2} \ge (s-a)(s-b)(s-c)$$

Now multiply by s on both side

$$\frac{s(abc)}{8} \ge s(s-a)(s-b)(s-c)$$

$$(a+b+c)\frac{abc}{16} \ge \Delta^2 \Rightarrow \frac{1}{4}\sqrt{(a+b+c)abc} \ge \Delta$$

Clearly equality occurs if, s − a = s − b = s − c

$\Rightarrow$ a = b = c

Hence Proved.

Q. **Prove that** $(r_3 + r_1)(r_3 + r_2)\sin C = 2r_3\sqrt{r_2r_3 + r_3r_1 + r_1r_2}$

Sol. LHS

$$= \left(\frac{\Delta}{s-c} + \frac{\Delta}{s-a}\right)\left(\frac{\Delta}{s-c} + \frac{\Delta}{s-b}\right)\sin C$$

$$= \Delta^2 \frac{b}{(s-c)(s-a)} \cdot \frac{a}{(s-c)(s-b)} . \sin C$$

$$= \Delta^2 . \frac{ab\sin C}{(s-c)^2(s-a)(s-b)} = \Delta^2 . \frac{2\Delta}{(s-a)(s-b)(s-c)^2}$$

$$= \frac{2\Delta}{(s-c)} \frac{\Delta^2}{(s-a)(s-b)(s-c)}$$

$$= 2r_3\sqrt{\frac{\Delta^2 \cdot \Delta^2}{(s-a)^2(s-b)^2(s-c)^2}}$$

$$= 2r_3\sqrt{\frac{\Delta^2 . s}{(s-a)(s-b)(s-c)}}$$

$$= 2r_3\sqrt{\Delta^2\left\{\frac{(s-a)+(s-b)+(s-c)}{(s-a)(s-b)(s-c)}\right\}}$$

$$= 2r_3\sqrt{\Delta^2\left\{\frac{1}{(s-b)(s-c)} + \frac{1}{(s-a)(s-c)} + \frac{1}{(s-a)(s-b)}\right\}}$$

$$= 2r_3\sqrt{r_2r_3 + r_1r_3 + r_1r_2}$$

Hence Proved.

Q. **If $r_1 = r + r_2 + r_3$ then prove that the triangle is a right angled triangle.**

Sol. Given equation is

$$\frac{\Delta}{s-a} = \frac{\Delta}{s} + \frac{\Delta}{s-b} + \frac{\Delta}{s-c}$$

$$\Rightarrow \frac{1}{s-a} - \frac{1}{s} = \frac{1}{s-b} + \frac{1}{s-c}$$

$$\Rightarrow \frac{s-(s-a)}{s(s-a)} = \frac{(s-c)+(s-b)}{(s-b)(s-c)}$$

$$\Rightarrow \frac{a}{s(s-a)} = \frac{a}{(s-b)(s-c)}$$

$\Rightarrow (s - b)(s - c) = s(s - a)$

$\Rightarrow s^2 - (b + c)s + bc = s^2 - as$

$(a - b - c)s + bc = 0$

$$\Rightarrow [a - (b + c)] \frac{\left[a + (b + c)\right]}{2} + bc = 0$$

$\Rightarrow a^2 - (b + c)^2 + 2bc = 0$

$\Rightarrow a^2 - b^2 - c^2 = 0$

$$\Rightarrow a^2 = b^2 + c^2 \Rightarrow \angle A = \frac{\pi}{2}$$

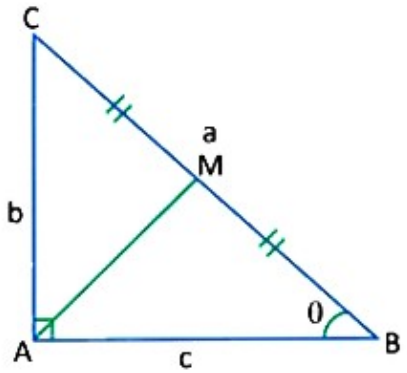
$\Rightarrow$ Right angled Δ

Q. **Given a right triangle with $\angle A = 90°$. Let M be the mid point of BC. If the inradii of the triangle ACM and ABM are r_1 and r_2 respectively then find the range of $\frac{r_1}{r_2}$.**

Sol. For ΔABC

$$\Delta = \frac{1}{2} AC.BC.\sin C$$

$$\Delta = \frac{1}{2} ab \sin C$$



for ΔAMC

area = $\Delta_1 = \frac{1}{2} AC.\ CM \sin C$

$$\Delta_1 = \frac{1}{2} b.\ \frac{a}{2} \sin C = \frac{\Delta}{2}$$

$$s_1 = AC + CM + AM \Rightarrow r_1 = \frac{\Delta_1}{s_1} = \frac{\Delta / 2}{s_1} \quad ...(1)$$

for ΔABM

area = $\Delta_2 = \frac{1}{2} AB.\ BM.\ \sin B = \frac{\Delta}{2}$

$$\text{and } s_2 = AB + BM + AM \Rightarrow r_2 = \frac{\Delta / 2}{s_2} \quad ...(2)$$

$$\frac{r_1}{r_2}=\frac{s_2}{s_1}=\frac{c+\frac{a}{2}+\frac{a}{2}}{b+\frac{a}{2}+\frac{a}{2}}=\frac{c+a}{b+a} \qquad ...(3)$$

Now let $c = a\cos\theta$, $b = a\sin\theta$ hence

$$\frac{r_1}{r_2}=\frac{1+\cos\theta}{1+\sin\theta} \Rightarrow \frac{r_1}{r_2}\bigg|_{max} < \frac{1+1}{1+0}=2$$

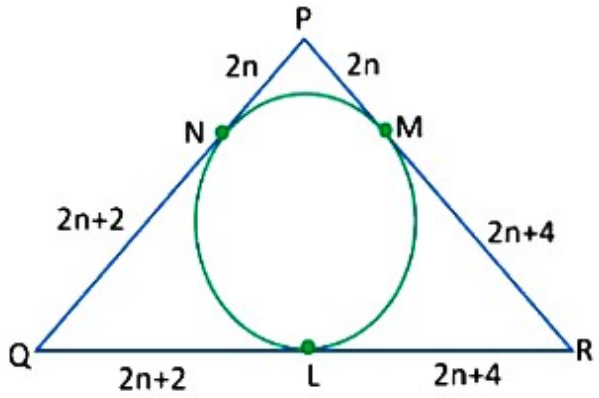
and $$\frac{r_1}{r_2}\bigg|_{min} > \frac{1+0}{1+1}=\frac{1}{2} \Rightarrow \frac{r_1}{r_2} \in \left(\frac{1}{2},2\right)$$

Q. **In a triangle PQR, P is the largest angle and $\cos P = \frac{1}{3}$. Further the incircle of the triangle touches the sides PQ, QR and RP at N, L and M respectively such that the lengths of PN, QL and RM are consecutive even integers. Then possible length(s) of side(s) of the triangle is/are**

(A) 16 **(B) 18** **(C) 24** **(D) 22**

Ans. **(B,D)**

Sol.



Let
PN = 2n = PM
QL = 2n + 2 = QN
RM = 2n + 4 = RL
Hence
PQ = 4n + 2
QR = 4n + 6, PR = 4n + 4
Now cosP = 1/3

$$\frac{PQ^2+PR^2-QR^2}{2PQ.PR}=\frac{1}{3}$$

$$\Rightarrow \quad \frac{2^2(2n+1)^2+2^2(2n+2)^2-2^2(2n+3)^2}{2.2(2n+1).2(2n+2)}=\frac{1}{3}$$

$\Rightarrow$ $3(4n^2 - 4) = 2(4n^2 + 6n + 2)$
$\Rightarrow$ $4n^2 - 12n - 16 = 0$
$\Rightarrow$ $n^2 - 3n - 4 = 0 \Rightarrow n = 4$
sides (PQ, QR, RP) = (18, 22, 20)

Q. **If the length of the perpendicular from the vertices of a triangle A, B, C on the opposite sides are p_1, p_2, p_3 then prove that $\frac{1}{p_1}+\frac{1}{p_2}+\frac{1}{p_3}=\frac{1}{r}=\frac{1}{r_1}+\frac{1}{r_2}+\frac{1}{r_3}$**

Sol. Area of triangle

$$\Delta=\frac{1}{2}a.p_1=\frac{1}{2}b.p_2=\frac{1}{2}c.p_3$$

$$\Rightarrow \quad \frac{1}{p_1}=\frac{a}{2\Delta},\frac{1}{p_2}=\frac{b}{2\Delta},\frac{1}{p_3}=\frac{c}{2\Delta}$$

$$\Rightarrow \quad \frac{1}{p_1}+\frac{1}{p_2}+\frac{1}{p_3}=\frac{1}{2\Delta}(a+b+c)=\frac{s}{\Delta}=\frac{1}{r}$$ **Hence Proved.**

$$\frac{1}{r_1}+\frac{1}{r_2}+\frac{1}{r_3}=\frac{s-a}{\Delta}+\frac{s-b}{\Delta}+\frac{s-c}{\Delta}$$

$$=\frac{3s-(a+b+c)}{\Delta}=\frac{s}{\Delta}=\frac{1}{r}$$ **Hence Proved.**

Q. **If P_1, P_2, P_3 are the altitudes of a triangle from the vertices A, B, C and Δ denotes the area of the triangle then prove that $\frac{1}{P_1}+\frac{1}{P_2}+\frac{1}{P_3}=\frac{2ab}{(a+b+c)}\cos^2\frac{C}{2}$.**

Sol. Area of triangle

$$\Delta=\frac{1}{2}a.P_1=\frac{1}{2}b.P_2=\frac{1}{2}c.P_3$$

$$\Rightarrow \frac{1}{P_1}=\frac{a}{2\Delta},\frac{1}{P_2}=\frac{b}{2\Delta},\frac{1}{P_3}=\frac{c}{2\Delta}$$

$$\Rightarrow \frac{1}{P_1}+\frac{1}{P_2}-\frac{1}{P_3}=\frac{1}{2\Delta}(a+b-c)$$

$$=\frac{((a+b)-c)((a+b)+c)}{2\Delta(a+b+c)}$$

$$=\frac{(a+b)^2-c^2}{2\Delta(a+b+c)}=\frac{2ab+(a^2+b^2-c^2)}{2\Delta(a+b+c)}$$

$$= \frac{2ab + 2abcosC}{2\Delta(a+b+c)} = \frac{2ab\left(2cos^2\frac{C}{2}\right)}{2\Delta(a+b+c)}$$

$$= \frac{2ab}{\Delta(a+b+c)}.cos^2\frac{c}{2}$$ **Hence Proved.**

Q. **If the sides of a triangle are in the ratio 3 : 7 : 8 then find R : r**

Sol. Let a = 3k, b = 7k, c = 8k then

$s = 9k$ and $\Delta = \sqrt{9k.6k.2k.k} = 6\sqrt{3}k^2$

$$R = \frac{abc}{4\Delta} = \frac{3k.7k.8k}{4.6\sqrt{3}k^2} = \frac{7}{\sqrt{3}}k$$

$$r = \frac{\Delta}{s} = \frac{6\sqrt{3}k^2}{9k^2} = \frac{2}{\sqrt{3}}k$$

Hence $\frac{R}{r} = \frac{7}{2}$

Centroid Of Triangle And Length Of Median:

If AD, BE and CF are medians of ΔABC then

$$AD = \frac{1}{2}\sqrt{2b^2 + 2c^2 - a^2}$$

$$BE = \frac{1}{2}\sqrt{2c^2 + 2a^2 - b^2}$$

$$CF = \frac{1}{2}\sqrt{2a^2 + 2b^2 - c^2}$$

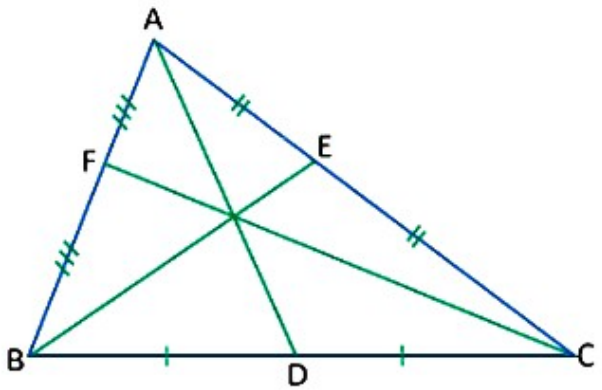
Proof:

In ΔABD

$$cosB = \frac{AB^2 + BD^2 - AD^2}{2AB.BD}$$

$$\Rightarrow \quad \frac{a^2 + c^2 - b^2}{2ac} = \frac{c^2 + \left(\frac{a}{2}\right)^2 - AD^2}{2.c.\frac{a}{2}}$$

$$\Rightarrow \quad a^2 + c^2 - b^2 = 2c^2 + \frac{a^2}{2} - 2AD^2$$



$\Rightarrow \quad 2AD^2 = c^2 - \frac{a^2}{2} + b^2$

$\Rightarrow \quad AD^2 = \frac{1}{4}\left(2b^2 + 2c^2 - a^2\right)$

$\Rightarrow \quad AD = \frac{1}{2}\sqrt{2b^2 + 2c^2 - a^2}$ **Hence Proved.**

Similarly, other sides can be proved.

Q. **In ΔABC, if lengths of medians BE and CF are 12 and 9 respectively. Find the maximum value of Δ.**

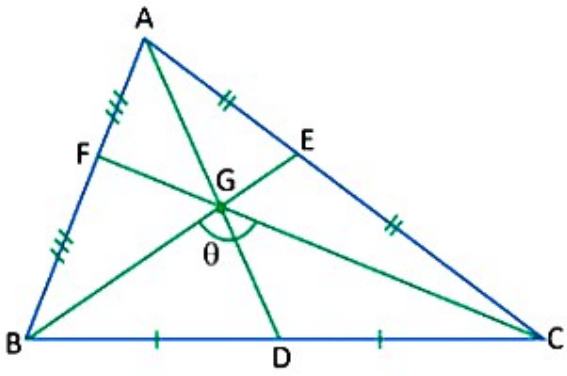
Sol. Given BE = 12, CF = 9
G is centroid of ΔABC
now

$BG = \frac{2}{3}$ BE = 8

$CG = \frac{2}{3}$ CF = 6

$ar(\Delta BGC) = \frac{1}{2}$ BG. CG. $\sin\theta$ = $24\sin\theta$

$ar(\Delta ABC) = 3.\ ar(\Delta BGC)$
$\Delta = 72\ \sin\theta$
hence $\Delta_{max} = 72$



Q. **In a non-right angle ΔPQR, let p, q, r denotes the lengths of the sides opposite to angles at P, Q, R respectively. The median from R meets the side PQ at S. The perpendicular from P meets the side QR at E, and RS and PE intersect at O. If $p = \sqrt{3}$, q = 1 and the radius of the circumcricle of the ΔPQR equal to 1 then which of the following options is/are correct?**

(A) Area of $\Delta SOE = \sqrt{3}$ / 12

(B) Radius of incircle of $\Delta PQR = \frac{\sqrt{3}\left(2-\sqrt{3}\right)}{2}$

(C) Length of RS = $\frac{\sqrt{7}}{2}$

(D) Length of OE = $\frac{1}{6}$

Ans. **(B,C,D)**

Sol. In ΔPQR

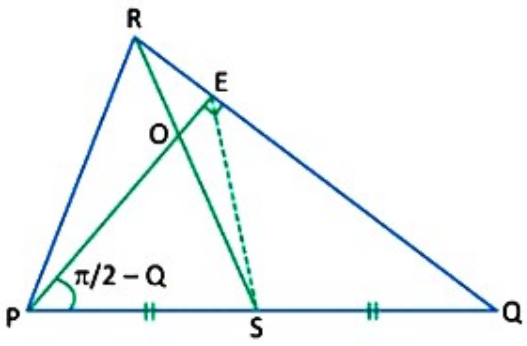
Circumradius

$$R' = \frac{q}{2\sin Q} = \frac{p}{2\sin P}$$

$$1 = \frac{1}{2\sin Q} = \frac{\sqrt{3}}{2\sin P}$$

$$\Rightarrow Q = \frac{\pi}{6}, P = \frac{2\pi}{3} \text{ and } R = \frac{\pi}{6} \qquad (\because \Delta PQR \text{ is not right angled})$$

Now $$\frac{r}{2\sin R} = 1 \Rightarrow r = 2\left(\frac{1}{2}\right) = 1$$



$\because \angle\theta = \angle R \Rightarrow \Delta PQR$ is isosceles

Hence PE is also a median $\Rightarrow$ O is centroid

Now length of median

$$PE = \frac{1}{2}\sqrt{2q^2 + 2r^2 - p^2} = \frac{1}{2}\sqrt{2 + 2 - 3} = \frac{1}{2}$$

$$RS = \frac{1}{2}\sqrt{2p^2 + 2q^2 - r^2} = \frac{1}{2}\sqrt{2(3) + 2 - 1} = \frac{\sqrt{7}}{2}$$

$$OE = \frac{1}{3}PE \Rightarrow OE = \frac{1}{6}$$

$$OS = \frac{2}{3}PS = \frac{2}{3}.\frac{\sqrt{7}}{2} = \frac{\sqrt{7}}{3}$$

$$ar(\Delta OES) = ar(\Delta PES) - ar(\Delta POS)$$

$$= \frac{1}{2}PE.PS.\sin\left(\frac{\pi}{2} - Q\right) - \frac{1}{2}.PO.PS\sin\left(\frac{\pi}{2} - Q\right)$$

$$= \frac{1}{2}\frac{1}{2}.\frac{1}{2}.\frac{\sqrt{3}}{2} - \frac{1}{2}\left(\frac{2}{3}.\frac{1}{2}\right).\frac{1}{2}.\frac{\sqrt{3}}{2}$$

$$= \frac{\sqrt{3}}{16} - \frac{1}{\sqrt{3}}.\frac{1}{8} = \frac{3-2}{16\sqrt{3}} = \frac{1}{16\sqrt{3}}$$

radius of incircle of ΔPQR

$$r' = \frac{\Delta}{s}$$

$$= \frac{\frac{1}{2}(p.q.\sin R)}{\frac{1}{2}(p + q + r)}$$

$$= \frac{\sqrt{3}.1\left(\frac{1}{2}\right)}{\left(\sqrt{3}+1+1\right)}$$

$$= \frac{\sqrt{3}}{2\left(2+\sqrt{3}\right)} \times \frac{\left(2-\sqrt{3}\right)}{\left(2-\sqrt{3}\right)}$$

$$\Rightarrow \quad r' = \frac{\sqrt{3}}{2}\left(2-\sqrt{3}\right)$$

Length Of Angle Bisector:

In ΔABC let AP, BQ and CR are angle bisectors then

$$AP = \left(\frac{2bc}{b+c}\right)\cos\left(\frac{A}{2}\right)$$; P is point where angle bisector of A meet BC

$$BQ = \left(\frac{2ac}{a+c}\right)\cos\left(\frac{B}{2}\right)$$

$$CR = \left(\frac{2ab}{a+b}\right)\cos\left(\frac{C}{2}\right)$$

Proof:

ar (ΔABC) = ar (ΔABP) + ar (ΔACP)

$$\frac{1}{2}\,bc\,\sin A = \frac{1}{2}\,c.AP\,\sin\frac{A}{2} + \frac{1}{2}b.AP\sin\frac{A}{2}$$

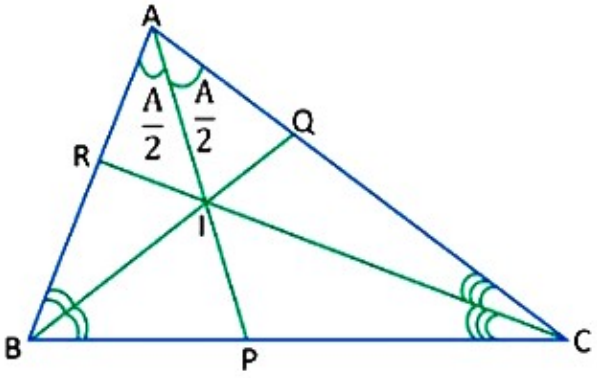
$$\Rightarrow \quad bc\,\sin A = (b+c)\,\sin\frac{A}{2}.AP$$

$$\Rightarrow \quad bc\left(2\sin\frac{A}{2}\cos\frac{A}{2}\right) = (b+c)\,\sin\frac{A}{2}.AP$$

$$\Rightarrow \quad 2bc\,\cos\frac{A}{2} = (b+c)\,.\,AP$$

$$\Rightarrow \quad AP = \frac{2bc}{b+c}\cos\frac{A}{2}$$

Similarly, others can be proved.



The Perimeter And Area Of A Regular Polygon Of N-sides Inscribed In A Circle

$\therefore$ AL = OA sin $\angle$AOL

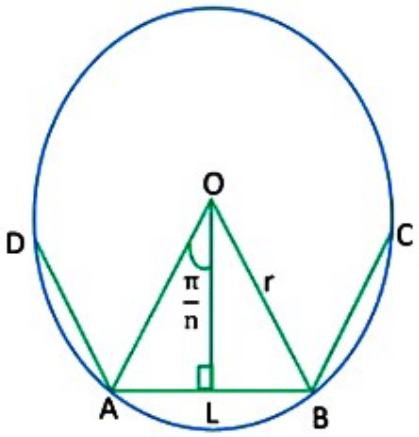
$$= r\sin\frac{\pi}{n}$$

Perimeter of polygon = nAB

$$= 2n.\ AL = 2nr \sin\frac{\pi}{n}$$

Area of polygon = n (Area of triangle AOB)

$$= \frac{nr^2}{2}\sin\frac{2\pi}{n}$$



The Perimeter And Area Of Regular Polygon Of N-sides Circumscribed About A Given Circle

Let DA, AB and BC be three successive sides of the polygon of n sides. Let r be the radius of given circle. OL is perpendicular to AB then

In ΔAOL, AL = OL tan $\frac{\pi}{n}$

$\therefore$ Perimeter of polygon = nAB = 2n AL

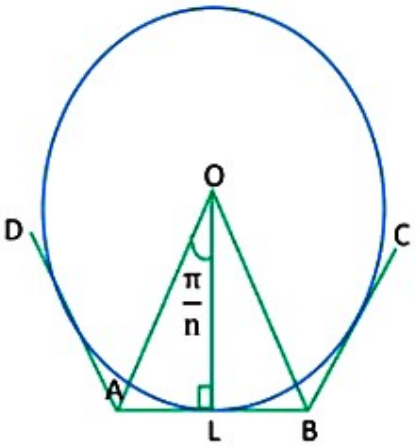
$$= 2n\ OL \tan\frac{\pi}{n}$$

$$= 2nr\ \tan\frac{\pi}{n}$$

Area of polygon = n (area of triangle AOB)

$$= n\frac{(OL.AB)}{2}$$

$$= nr^2\tan\frac{\pi}{n}$$



Distance of orthocenter from vertices

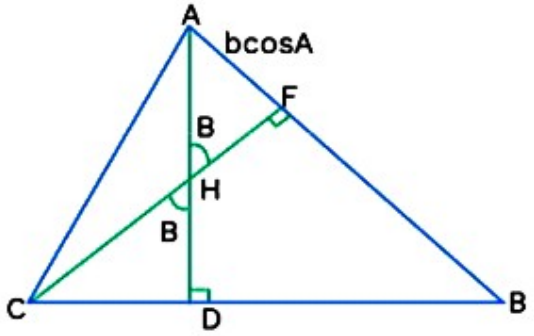
$\Rightarrow$ AH = $\frac{b\cos A}{\sin B}$

$$= 2R\cos A$$

where H is the orthocenter of triangle ABC, similarly

BH = 2R cosB

CH = 2R cosC



Distance of orthocenter from sides

DH = CH. cosB

$\Rightarrow$ DH = 2R cosB cosC

where D is foot of altitude from A

Q. **Given (AH) (BH) (CH) = 3 and $AH^2 + BH^2 + CH^2 = 7$ (where H is orthocenter of ΔABC) Find**

(a) the ratio $\frac{\prod \cos A}{\sum \cos^2 A}$

Sol. $\because$ $2R\cos A.\ 2R\cos B.\ 2R\cos C = 3$

$\Rightarrow \cos A.\cos B.\cos C = \frac{3}{8R^3}$ ...(1)

also $(2R\cos A)^2 + (2R\cos B)^2 + (2R\cos C)^2 = 7$

$4R^2(\cos^2 A + \cos^2 B + \cos^2 C) = 7$

$\cos^2 A + \cos^2 B + \cos^2 C = \frac{7}{4R^2}$ ...(2)

$$\frac{1+\cos 2A}{2} + \frac{1+\cos 2B}{2} + \frac{1+\cos 2C}{2} = \frac{7}{4R^2}$$

$$3 + (\cos 2A + \cos 2B + \cos 2C) = \frac{7}{2R^2}$$

$$3 + (-1 - 4\cos A.\cos B.\cos C) = \frac{7}{2R^2}$$

$$2 - 4.\frac{3}{8R^3} = \frac{7}{2R^2} \qquad \{\text{from (1)}\}$$

$\Rightarrow 4R^3 - 3 = 7R$

$\Rightarrow 4R^3 - 7R - 3 = 0$

$\Rightarrow (R+1)(4R^2 - 4R - 3) = 0$

$\Rightarrow (R+1)(4R^2 - 6R + 2R - 3) = 0$

$\Rightarrow (R+1)(2R+1)(2R-3) = 0 \Rightarrow R = \frac{3}{2}$

now dividing (1) by (2),

$$\frac{\prod \cos A}{\sum \cos^2 A} = \frac{3}{8R^3} \times \frac{4R^2}{7} = \frac{3}{14R} = \frac{3}{14\left(\frac{3}{2}\right)} = \frac{1}{7}$$

Q. **(b) the product HD. HE. HF**

Sol. $(2R\cos B\cos C)(2R\cos A\cos C)(2R\cos A\cos B)$

$= 8R^3\cos^2 A.\cos^2 B.\cos^2 C$

$= 8R^3.\left(\frac{3}{8R^3}\right)^2$ (From equation (1))

$$= \frac{9}{8R^3} = \frac{9}{8\left(\frac{3}{2}\right)^3} = \frac{1}{3}$$

Q. **(c) the value of R**

Sol. $R = \frac{3}{2}$

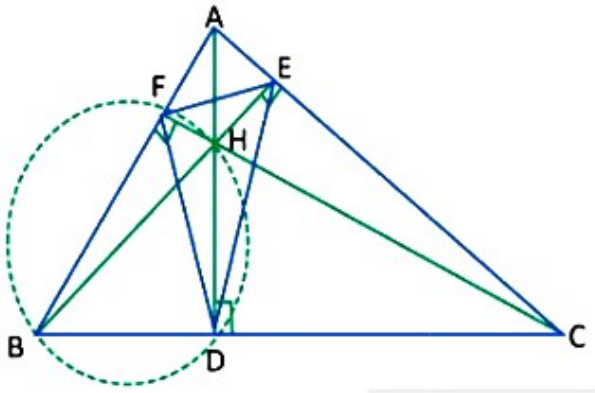
Pedal Triangle

In an acute-angled triangle, orthocenter of ΔABC is the incentre of the Pedal triangle DEF.

Sides and angle of pedal triangle in acute angled triangle

Sides: $EF = a\cos A$
$DF = b\cos B$
$DE = c\cos C$

Angles: $\angle FDE = \pi - 2A$
$\angle DEF = \pi - 2B$
$\angle EFD = \pi - 2C$



Circumradius of pedal triangle $= \frac{R}{2}$

(where R is circumradius of ΔABC)

Proof: In ΔAEF

$$\cos A = \frac{AE^2 + AF^2 - EF^2}{2.AE.AF}$$

$$\Rightarrow \quad \cos A = \frac{(c\cos A)^2 + (b\cos A)^2 - EF^2}{2(c\cos A)(b\cos A)}$$

$\Rightarrow \quad 2bc\cos^3 A = (b^2 + c^2)\cos^2 A - EF^2$

$\Rightarrow \quad EF^2 = (b^2 + c^2)\cos^2 A - 2bc\cos^3 A$

$\Rightarrow \quad EF^2 = \cos^2 A\{b^2 + c^2 - 2bc\cos A\}$

$\Rightarrow \quad EF^2 = \cos^2 A\{b^2 + c^2 - (b^2 + c^2 - a^2)\}$

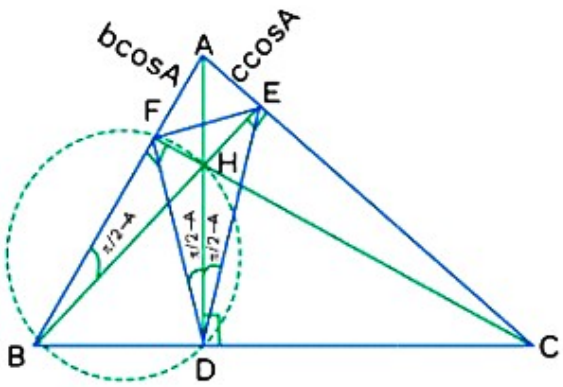
$\Rightarrow \quad EF^2 = \cos^2 A.\, a^2$

$\Rightarrow \quad EF = a\cos A$

Now $\angle FDE = \angle FDA + \angle ADE$

$$= \frac{\pi}{2} - A + \frac{\pi}{2} - A$$

$$= \pi - 2A$$



Q. **Prove that circumradius of ΔHBC, ΔHCA, ΔHAB, ΔABC are same.**

Sol. Circumradius of

(i) $\Delta ABC \Rightarrow R = \frac{a}{2\sin A} = \frac{b}{2\sin B} = \frac{c}{2\sin C}$ ...(1)

(ii) $\Delta HBC \Rightarrow R_1 = \frac{BC}{2\sin(BHC)} = \frac{a}{2\sin(B+C)}$

$= \frac{a}{2\sin(\pi - A)} = \frac{a}{2\sin A}$ ...(2)

(iii) $\Delta HCA \Rightarrow R_2 = \frac{CA}{2\sin(CHA)} = \frac{b}{2\sin(C+A)}$

$= \frac{b}{2\sin(\pi - B)} = \frac{b}{2\sin B}$ ...(3)

(iv) $\Delta HAB \Rightarrow R_3 = \frac{AB}{2\sin(AHB)} = \frac{c}{2\sin(A+B)}$

$= \frac{c}{2\sin(\pi - C)} = \frac{c}{2\sin C}$ ...(4)

by all the four equations
$R = R_1 = R_2 = R_3$

Q. **Prove that**

(i) $\Delta_p = 2\Delta \cos A \cos B \cos C$; Δ_p is area of pedal triangle.

(ii) $R_p = \frac{R}{2}$

(iii) $r_p = 2R \cos A \cos B \cos C$

Sol. **(i)** $\Delta_p = \frac{1}{2}$ DE. EF sin(DEF)

$= \frac{1}{2}$ (c cosC) (a cosA) sin(π − 2B)

$= \frac{1}{2}$ ac cos C cosA. (2sinB cosB)

$= 2\Delta \cos A \cos B \cos C \left\{\because \Delta = \frac{1}{2} ac \sin B\right\}$

(ii) $R_p = \dfrac{EF}{2\sin(FDE)} = \dfrac{a\cos A}{2\sin(\pi - 2A)}$

$= \dfrac{a\cos A}{2(2\sin A\cos A)} = \dfrac{1}{2}.\dfrac{a}{2\sin A}$

$= \dfrac{R}{2}$

(iii) $r_p = 4R_p \sin\left(\dfrac{\pi - 2A}{2}\right)\sin\left(\dfrac{\pi - 2B}{2}\right)\sin\left(\dfrac{\pi - 2C}{2}\right)$

$= 4\dfrac{R}{2}.\sin\left(\dfrac{\pi}{2} - A\right)\sin\left(\dfrac{\pi}{2} - B\right)\sin\left(\dfrac{\pi}{2} - C\right)$

$= 2R\cos A\cos B\cos C$ **Hence Proved.**

Q **Area $\Delta ABC = 18$; Area $\Delta BDF = 2$, length of side DF = $2\sqrt{2}$ then find R (where D, F are feet of altitude from A and C respectively)**

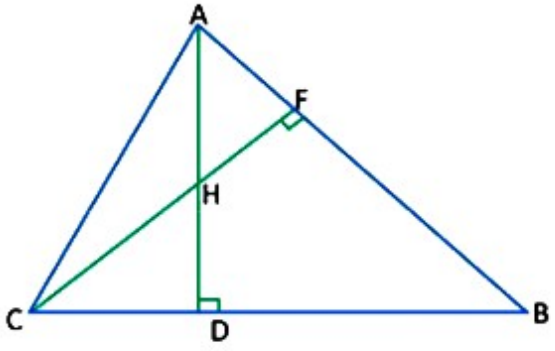
Sol. Given $\Delta = 18$

$ar(\Delta BDF) = 2$

$\dfrac{1}{2}$ FB. DB sin(FBD) = 2

$\dfrac{1}{2}$(a cosB) (c cosB) sinB = 2

$\Rightarrow \left(\dfrac{1}{2}ac\sin B\right).\cos^2 B = 2 \Rightarrow 18\cos^2 B = 2$



$\Rightarrow \cos B = \dfrac{1}{3}$

also DF is one of the side of pedal triangle of ΔABC hence

$DF = b\cos B = 2\sqrt{2}$

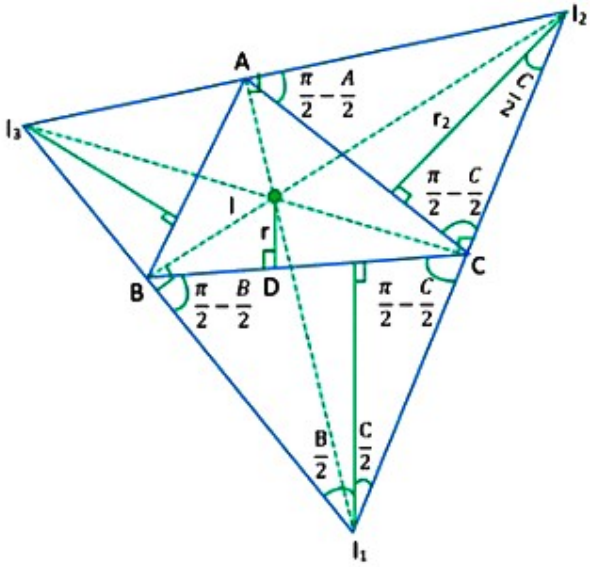
$\Rightarrow b\left(\dfrac{1}{3}\right) = 2\sqrt{2} \Rightarrow b = 6\sqrt{2}$

now $R = \dfrac{b}{2\sin B} = \dfrac{6\sqrt{2}}{2\sqrt{1 - \frac{1}{9}}} = \dfrac{6\sqrt{2}}{2.\left(\frac{2\sqrt{2}}{3}\right)}$

$= \dfrac{9}{2}$ units

External Or Ex-centric Triangle

Triangle is formed by joining the 3 excentres is called excentral triangle or excentric triangle ΔABC is a pedal triangle for $\Delta I_1I_2I_3$ therefore incentre of ΔABC is orthocenter for $\Delta I_1I_2I_3$.



Note:

(i) Internal angles are $\frac{\pi}{2}-\frac{A}{2},\frac{\pi}{2}-\frac{B}{2},\frac{\pi}{2}-\frac{C}{2}$

(ii) Sides of excentric Δ are

$$I_1I_2 = 4R\cos\frac{C}{2}$$

$$I_2I_3 = 4R\cos\frac{A}{2}$$

$$I_1I_3 = 4R\cos\frac{B}{2}$$

Distance between Incentre and excentre:

$$II_1 = 4R\sin\frac{A}{2}$$

$$II_2 = 4R\sin\frac{B}{2}$$

$$II_3 = 4R\sin\frac{C}{2}$$

Proof: In ΔIBD

$$\sin\frac{B}{2}=\frac{ID}{BI} \Rightarrow BI = \frac{r}{\sin\frac{B}{2}} \qquad ...(1)$$

In ΔIBI_1

$$\cos(BII_1) = \frac{BI}{II_1}$$

$$\Rightarrow \quad \cos\left(\frac{\pi}{2}-\frac{c}{2}\right)=\frac{\frac{r}{\sin\frac{B}{2}}}{II_1}$$

$\{\angle BII_1 = \angle BCI_1 \quad \because B, I_1, C, I \text{ are concyclic}\}$

$$\Rightarrow \quad \sin\frac{C}{2}=\frac{r}{II_1\sin\frac{B}{2}}$$

$$\Rightarrow \quad II_1 = \frac{r}{\sin\frac{B}{2}.\sin\frac{C}{2}} = 4R\sin\frac{A}{2}$$ **Hence Proved.**

Now in ΔII_1I_2 by sine rule

$$\frac{I_1I_2}{\sin(I_1II_2)} = \frac{II_1}{\sin\frac{A}{2}}$$

$$\Rightarrow \quad \frac{I_1I_2}{\sin\left(\pi - \frac{A+B}{2}\right)} = \frac{4R\sin\frac{A}{2}}{\sin\frac{A}{2}}$$

{$\because$ I, C, I_2, A are concyclic}

$$\Rightarrow \quad I_1I_2 = 4R\sin\left(\frac{A+B}{2}\right)$$

$$\Rightarrow \quad I_1I_2 = 4R\cos\frac{C}{2}$$ **Hence Proved.**

Distance between special points:

(A) The distance between the circumcenter and orthocenter of triangle ABC is

$= R\sqrt{1-8\cos A\cos B\cos C}$

Proof: O $\rightarrow$ Circumcenter

H $\rightarrow$ Orthocenter

In ΔAOH

$$\cos\theta = \frac{OA^2 + AH^2 - OH^2}{2OA.AH}$$

$$\cos\{A - (\pi - 2C)\} = \frac{R^2 + (2R\cos A)^2 - OH^2}{2.R.2R\cos A}$$

$$\cos(C - B) = \frac{R^2 + 4R^2\cos^2A - OH^2}{4R^2\cos A}$$

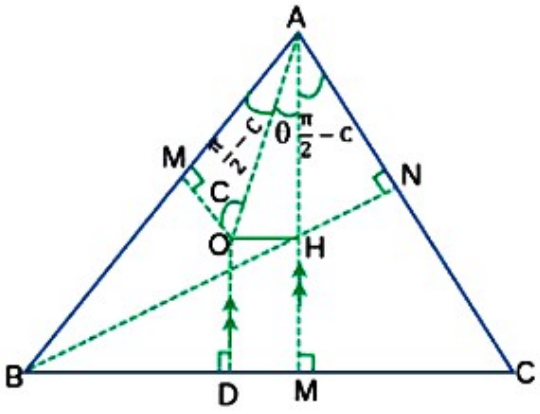
$4R^2\cos A.\cos(C - B) = R^2 + 4R^2\cos^2A - OH^2$

$\Rightarrow \quad OH^2 = R^2 + 4R^2\cos A\{\cos A - \cos(C - B)\}$

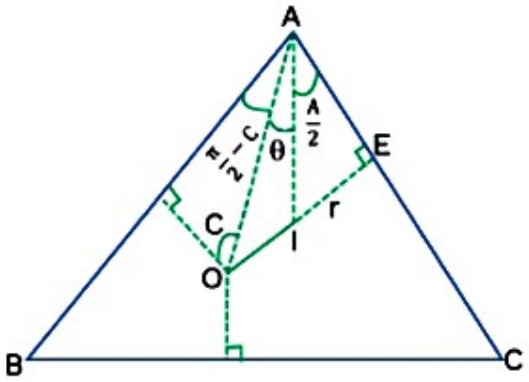
$OH^2 = R^2 - 4R^2\cos A\{\cos(B + C) + \cos(C - B)\}$

$OH^2 = R^2 - 4R^2\cos A(2\cos B.\cos C)$

$\Rightarrow \quad OH = R\sqrt{1-8\cos A\cos B\cos C}$ **Hence Proved.**



(B) Distance between incentre & circumcenter is given by $\sqrt{R^2 - 2Rr}$



Proof: O → Circumcenter
I → Incenter
In ΔAOI

$$\cos\theta = \frac{OA^2 + IA^2 - OI^2}{2.OA.IA}$$

$$\Rightarrow \quad OI^2 = OA^2 + IA^2 - 2OA.\,IA\cos\theta$$

$$= R^2 + \left(\frac{r}{\sin\frac{A}{2}}\right)^2 - 2R.\left(\frac{r}{\sin\frac{A}{2}}\right)\cos\left(\frac{A}{2} - \frac{\pi}{2} + C\right)$$

$$= R^2 + \left(4R\sin\frac{B}{2}\sin\frac{C}{2}\right) - 2R.4R\sin\frac{B}{2}\sin\frac{C}{2}\cos\left(\frac{C-B}{2}\right)$$

$$= R^2 + 8R^2\sin\frac{B}{2}\sin\frac{C}{2}\left\{2\sin\frac{B}{2}\sin\frac{C}{2} - \cos\left(\frac{C-B}{2}\right)\right\}$$

$$= R^2 + 8R^2\sin\frac{B}{2}\sin\frac{C}{2}\left\{-\cos\left(\frac{B+C}{2}\right)\right\}$$

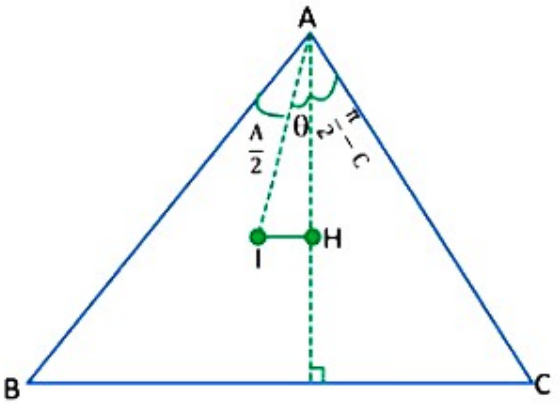
$$= R^2 - 8R^2\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2}$$

$$= R^2 - 2Rr$$

$$\Rightarrow OI = \sqrt{R^2 - 2Rr}$$

Hence Proved.

(C) Distance between incenter and orthocenter is $\sqrt{2r^2 - 4R^2\cos A\cos B\cos C}$



Proof: I → incentre
H → orthocenter

$$AI = 4R\sin\frac{B}{2}\sin\frac{C}{2}$$

$AH = 2R\cos A$
Now in ΔAIH,

$$\cos\theta = \frac{AI^2 + AH^2 - IH^2}{2AI.AH}$$

$$\Rightarrow IH^2 = AI^2 + AH^2 - 2AI.AH\cos\theta$$

$$= 16R^2\sin^2\frac{B}{2}\sin^2\frac{C}{2} + 4R^2\cos^2A - 2.4R\sin\frac{B}{2}\sin\frac{C}{2}.2R\cos A.\cos\left(\frac{C-B}{2}\right)$$

$$= 4R^2\cos^2A + 16R^2\sin^2\frac{B}{2}\sin^2\frac{C}{2} - 16R^2\cos A\sin\frac{B}{2}\sin\frac{C}{2}\left\{\cos\frac{C}{2}\cos\frac{B}{2} + \sin\frac{C}{2}\sin\frac{B}{2}\right\}$$

$$= 4R^2\cos^2A + 16R^2\sin^2\frac{B}{2}\sin^2\frac{C}{2} - 4R^2\cos A\sin B\sin C - 16R^2\cos A\sin^2\frac{B}{2}\sin^2\frac{C}{2}$$

$$= 4R^2\cos A\{\cos A - \sin B\sin C\} + 16R^2\sin^2\frac{B}{2}\sin^2\frac{C}{2}(1-\cos A)$$

$$= 4R^2\cos A(-\cos B\cos C) + 16R^2\sin^2\frac{B}{2}\sin^2\frac{C}{2}2\sin^2\frac{A}{2}$$

$$= -4R^2\cos A\cos B\cos C + 2\left\{4R\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2}\right\}^2$$

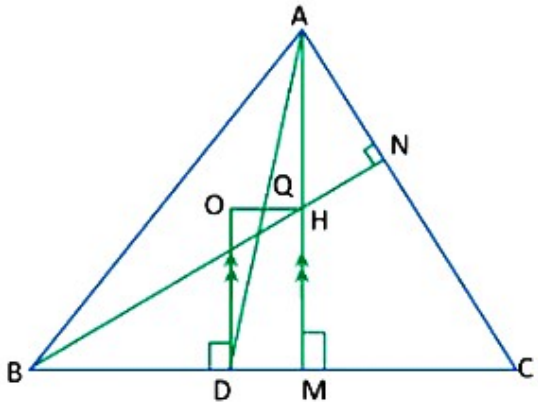
$$= -4R^2\cos A\cos B\cos C + 2r^2$$

$$\Rightarrow IH = \sqrt{2r^2 - 4R^2\cos A\cos B\cos C}$$

Hence Proved.

Q. **Prove that in any triangle H, G, O are collinear and G divides the line HO in 2 : 1 where H is orthocenter, O is circumcenter and G is centroid.**

Sol. Let O → Circumcentre
H → Orthocentre
and OD ∥ HM
also Q is point of intersection of AD and OH
Now In ΔAQH & ΔDQO
∠DQO = ∠AQH (Vertically opposite)
∠ODQ = ∠HAQ (alternate)
⇒ ΔDOQ ~ ΔAHQ
Now OD = RcosA
AH = 2RcosA



By similarity $\frac{AQ}{QD} = \frac{HQ}{QO} = \frac{AH}{OD} = \frac{2R\cos A}{R\cos A}$

$$\Rightarrow \frac{AQ}{QD} = \frac{HQ}{QO} = \frac{2}{1}$$

Also, we know D is mid point of BC (As OD is ⊥ bisector)

⇒ AD is median and Q is such that $\frac{AQ}{QD} = \frac{2}{1}$ ⇒ Q = G(centroid)

which implies H, G, O are collinear and $\frac{HG}{GO} = \frac{2}{1}$

Hence Proved.

Q. **Prove that: $(OG)^2 = R^2 - \frac{1}{9}(a^2 + b^2 + c^2)$ where O is circumcenter & G is centroid**

Sol. $OG = \frac{1}{3} OH = \frac{1}{3}R\sqrt{1 - 8\cos A\cos B\cos C}$

$$(OG)^2 = \frac{1}{9}R^2\left(1 - 4\left\{2\cos A\cos B\right\}\cos C\right)$$

$$= \frac{1}{9}R^2\ (1 - 4\{\cos(A + B) + \cos(A-B)\}\cos C)$$

$$= \frac{1}{9}R^2\ (1 - 4\{-\cos C + \cos(A - B)\}.\cos C)$$

$$= \frac{1}{9}R^2\{1 + 4\cos^2 C + 4\cos(A - B)\cos(A + B)\}$$

$$= \frac{1}{9}R^2\{1 + 4(1 - \sin^2 C) + 4\cos^2 A - 4\sin^2 B\}$$

$$= \frac{1}{9}R^2\{5 - 4\sin^2 C + 4(1 - \sin^2 A) - 4\sin^2 B\}$$

$$= \frac{1}{9}R^2\{9 - (2\sin A)^2 - (2\sin B)^2 - (2\sin C)^2\}$$

$$= \frac{1}{9}R^2\left\{9 - \frac{a^2}{R^2} - \frac{b^2}{R^2} - \frac{c^2}{R^2}\right\} = R^2 - \frac{1}{9}\left(a^2 + b^2 + c^2\right)$$ **Hence Proved.**

Ambiguous cases:

Q. **Two sides of a triangle are of lengths $\sqrt{6}$ and 4 and the angle opposite to smaller side is 30°. How many such triangles are possible? Find the length of their third side and area.**

Sol. Let $a = \sqrt{6}$, $b = 4$ and $A = 30°$ $(\because a < b)$

Now $\frac{a}{\sin A} = \frac{b}{\sin B}$

$$\Rightarrow \frac{\sqrt{6}}{\left(\frac{1}{2}\right)} = \frac{4}{\sin B} \Rightarrow \sin B = \sqrt{\frac{2}{3}}$$

Now two values of B are possible hence two triangles are possible

Now $\frac{b}{\sin B} = \frac{c}{\sin C}$

$c = \frac{\sin(A+B)}{\sin B}.b$

$c = \left\{\frac{\sin A.\cos B}{\sin B} + \frac{\cos A \sin B}{\sin B}\right\} b$

$= (\sin A \cot B + \cos A)b$

$= \left\{\frac{1}{2}\left(\pm\frac{1}{\sqrt{2}}\right) + \frac{\sqrt{3}}{2}\right\} 4 = \left(\sqrt{6} \pm 1\right)\sqrt{2}$ unit

Area $= \frac{1}{2}$ bc sinA

$= \frac{1}{2} 4\left(\sqrt{6} \pm 1\right)\sqrt{2} \times \frac{1}{2} = \left(\sqrt{6} \pm 1\right)\sqrt{2}$ sq. unit

Q. **If b = 3, c = 4 and B = $\frac{\pi}{3}$, then find number of triangles that can be constructed.**

Sol. $\frac{b}{\sin B} = \frac{c}{\sin C}$

$\Rightarrow \frac{3}{\left(\frac{\sqrt{3}}{2}\right)} = \frac{4}{\sin C} \Rightarrow \sin C = \frac{2}{\sqrt{3}} > 1$

$\Rightarrow$ No such triangle is possible.

Q. **If b = 5, c = $5\sqrt{2}$ and B = 45° then find the number of triangles that can be constructed.**

Sol. $\frac{b}{\sin B} = \frac{c}{\sin C}$

$\Rightarrow \frac{5}{\left(\frac{1}{\sqrt{2}}\right)} = \frac{5\sqrt{2}}{\sin C} \quad \Rightarrow \sin C = 1$

$C = \frac{\pi}{2}$

Hence only one triangle is possible

Q. **If b = 3, $c = \frac{5}{2}$ and C = 30° then find the number of triangles that can be constructed.**

Sol. $\frac{b}{\sin B} = \frac{c}{\sin C}$

$\Rightarrow \frac{3}{\sin B} = \frac{\frac{5}{2}}{\left(\frac{1}{2}\right)} \Rightarrow \sin B = \frac{3}{5}$

Two values of B are possible, hence two such triangles are possible

Q. **If A = 30°, a = 7 and b = 8 then find the number of triangles that can be constructed.**

Sol. $\frac{a}{\sin A} = \frac{b}{\sin B}$

$\Rightarrow \frac{7}{\left(\frac{1}{2}\right)} = \frac{8}{\sin B} \Rightarrow \sin B = \frac{4}{7}$

Two values of B are possible, hence two such triangles are possible.

Nine Point Circle:

It is a circle that can be constructed for any given triangle. It so named because it passes through nine significant concyclic points defined from the triangle. These nine points are

- The midpoints of each side of triangle
- The foot of each altitude
- The midpoint of the line segment from each vertex of the triangle to the orthocenter.

